UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Total Derivative, Chain Rule and Implicit Functions

Complete study pack: composite functions, the total derivative and total differential, rates of change, the chain rule for functions of several variables, and implicit differentiation in two and three variables

Compiled from the PESU class 003 and 004 slide decks, the four class 003–004 AV-summary transcripts, the Unit 1 class-work sheet and past ESA papers. Generated 14 October 2026.

· How to use this pack

Topics 1–7 follow the order of the lectures: class 003 covers topics 1–5, class 004 topics 5–7. Several problems on the slides contain errors; they are corrected here and marked as exam traps. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·7 · S004·9	Deck 003 / 004, slide number (rendered .pptx)
T003a 00:06:27	Transcript 003 part I: “Total derivative, chain rule and implicit functions”
T003b 00:01:10	Transcript 003 part II: “Problems on total derivatives, chain rule”
T004a 00:13:09	Transcript 004 part I: “Implicit functions and problems”
T004b 00:02:31	Transcript 004 part II: “Problems on composite and implicit functions”
CW U1 P14 · ESA Jan-24 s3 1b	Unit 1 class-work problem 14 · past-paper question

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1 Composite functions and the one-variable chain rule

A composite function is a function of a function, such as $y = e^{\sin(x^2)}$. You differentiate it from the outside in: the outer function first, then the next one, then the innermost, and multiply. Every rule in this pack is this same idea extended to several variables.

Key terms and definitions

Composite function. A function of a function: $f(g(h(x)))$, or a function of several variables whose variables are themselves functions of other variables. S003·4 · T003a 00:00:59

Formulas

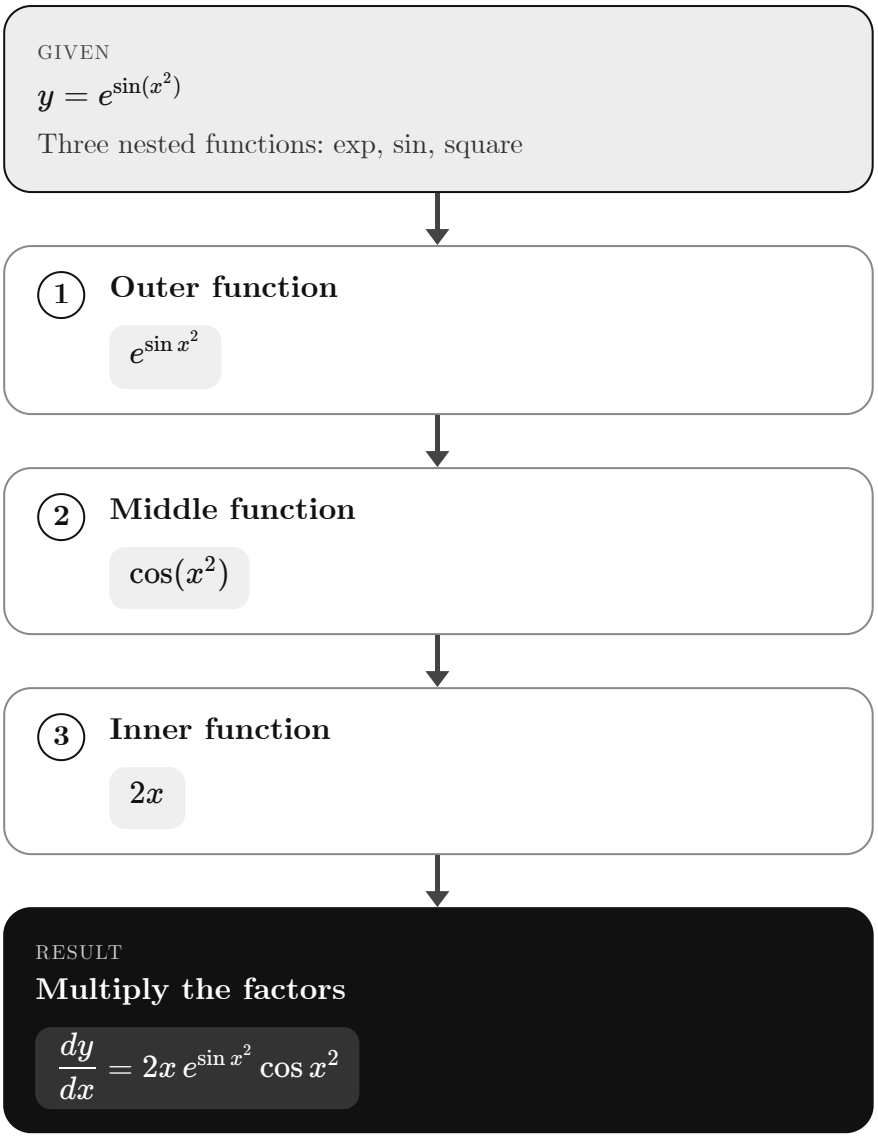
E1.1 · Chain rule, one variable ESA ★

$$\frac{d}{dx} f\big(g(h(x))\big) = f'\big(g(h(x))\big) \cdot g'\big(h(x)\big) \cdot h'(x)$$

GIVES	Derivative of a composition: outer to inner, multiplied.
APPLIES	Each function differentiable.
NOTE	Example: $\frac{d}{dx} e^{\sin x^2} = e^{\sin x^2} \cos x^2 \cdot 2x$.

S003·4 · T003a 00:00:33

Flowcharts



Remark. The multivariable rules in this pack work the same way, except that each intermediate variable adds its own term.

Figure F1.1. Differentiating a composite function from the outside in. ESA ★

S003·4 · T003a 00:00:33

Important points and why

1. For one variable you differentiate a composite function from the outside in: f first, then g, then h. ESA ★ T003a 00:01:19
2. For several variables the first question is always "differentiate with respect to what?", because there is more than one variable. ESA ★ T003a 00:01:45

Worked examples

W1.1 · Outside-in differentiation ESA ★

given $y = e^{\sin(x^2)}$.

step 1 Outer: $\dfrac{d}{d(\sin x^2)} e^{\sin x^2} = e^{\sin x^2}$.

step 2 Middle: $\dfrac{d}{d(x^2)} \sin(x^2) = \cos(x^2)$.

step 3 Inner: $\dfrac{d}{dx} x^2 = 2x$.

step 4 Multiply the three factors.

answer $\boxed{\dfrac{dy}{dx} = 2x\, e^{\sin x^2} \cos x^2}$

check sympy gives the same.

S003·4 · T003a 00:00:33

short-note recap

A composite function is a function of a function, e.g. $y = e^{\sin x^2}$. Its derivative is found from the outside in and the factors are multiplied: $e^{\sin x^2} \cdot \cos x^2 \cdot 2x$. For several variables the same idea gives the total derivative and the chain rule.

2 The total derivative rule

If $z = f(x, y)$ and both x and y are functions of one variable t, then z is ultimately a function of t alone. Its ordinary derivative dz/dt is the total derivative. It adds one term per path from z to t: $z_x\,dx/dt + z_y\,dy/dt$. A tree diagram shows the paths.

Key terms and definitions

Total derivative. If $z = f(x, y)$ where x and y are functions of one independent variable t, the ordinary derivative dz/dt is the total derivative of z with respect to t. S003·5 · T003b 00:00:52

Tree diagram. A diagram with z at the root, the intermediate variables in the middle and the final variable(s) at the leaves. Each branch is labelled with a derivative; multiply along each path and add over the paths. S003·5 · S003·14 · T003a 00:19:22 · T003a 00:25:02

> *"If $z = f(x, y)$ where x and y are functions of one independent variable 't' then we can find the derivative of z with respect t as* $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$*."* S003·5

Formulas

E2.1 · Total derivative rule ESA ★★

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

GIVES	Rate of change of z with respect to the single variable t.
APPLIES	$z = f(x, y)$, $x = x(t)$, $y = y(t)$.
NOTE	Ordinary d on z, x, y with respect to t; partial ∂ only on z with respect to x, y.

S003·5 · T003a 00:05:54 · T003b 00:01:10

E2.2 · Total derivative, three variables ESA ★★

$$\frac{df}{dt} = f_x\frac{dx}{dt} + f_y\frac{dy}{dt} + f_z\frac{dz}{dt}$$

GIVES	One term per intermediate variable.
APPLIES	$f = f(x, y, z)$, each of x, y, z a function of t.
NOTE	Used in CW U1 P11 and in the triangle problem with x, y, θ.

ADDED FOR CLARITY S003·9 · CW U1 P11

Graphs and diagrams

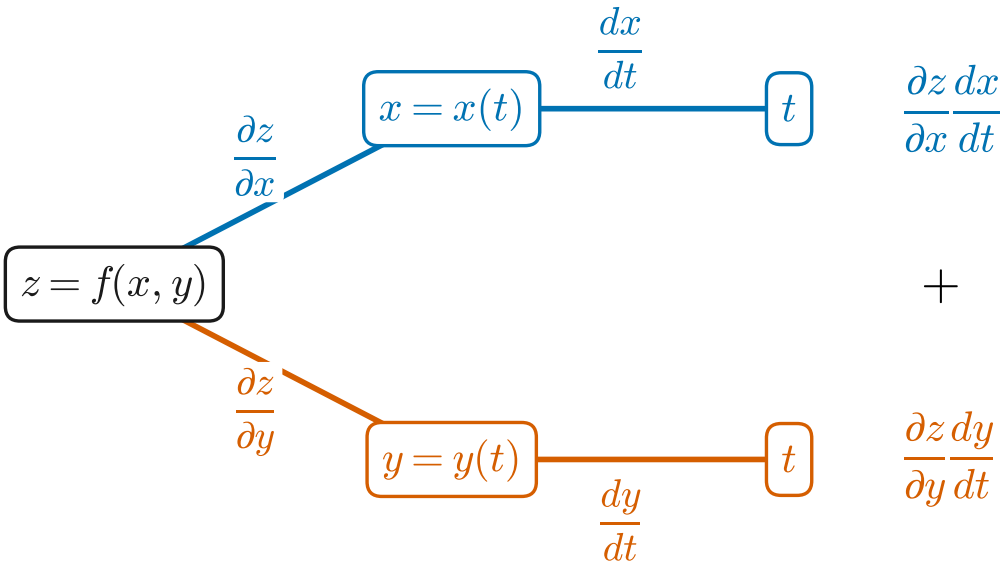


Figure G2.1. Tree diagram for the total derivative. ESA ★★

S003·5 · T003a 00:19:22

How to read it. Start at z. Each branch carries the derivative of the upper variable with respect to the lower one. Multiply along each path from z down to t, then add the two paths: $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$. In an exam, draw it first so that no path is missed.

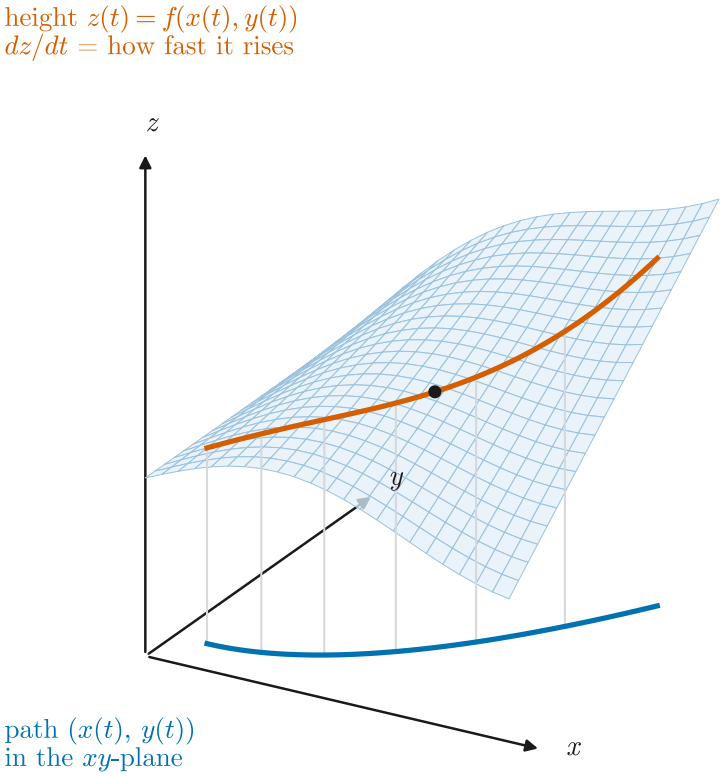
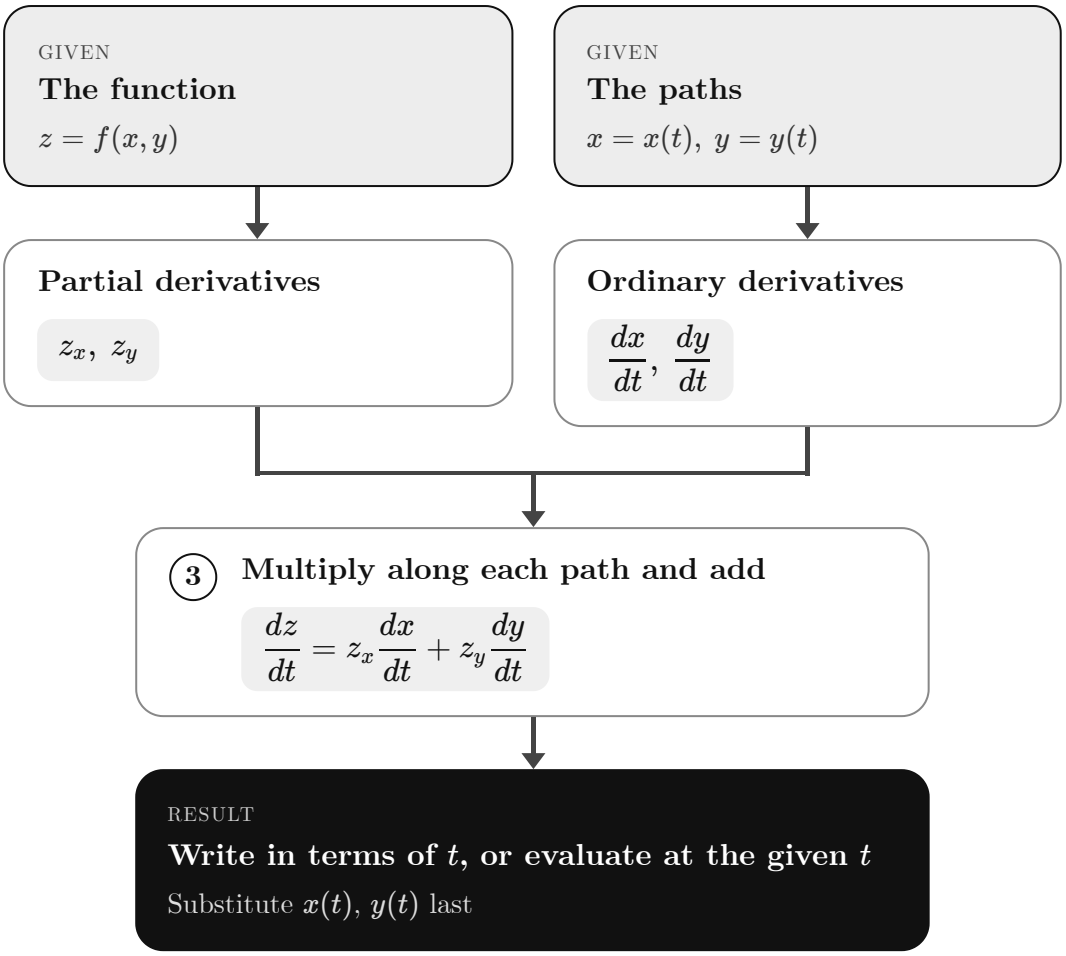


Figure G2.2. The total derivative as the slope along a path. ESA ★

ADDED FOR CLARITY S003·5 · T003a 00:05:20

How to read it. As t runs, the point $(x(t), y(t))$ moves along the blue path in the xy-plane. The vermilion curve is the surface height above it, $z(t) = f(x(t), y(t))$. dz/dt is how fast that height changes as t increases; it combines the slopes z_x and z_y weighted by how fast x and y move. The surface is an illustrative choice.

Flowcharts



Remark. Check on a small example by substituting first: both routes must agree. On slide 003·6 they do not, which exposes the slide's error (W2.1).

Figure F2.1. Total derivative in three moves. ESA ★★

S003·5 · S003·18 · T003b 00:01:10

Important points and why

1. You can always substitute $x(t)$ and $y(t)$ into z and differentiate directly, but this is often messy. The total derivative rule avoids substituting first. ESA ★★ T003a 00:02:39 · T003a 00:03:13
2. Both methods must give the same answer; substituting first is a good check on a short example. ESA ★★ ADDED FOR CLARITY T003a 00:04:35
3. After using the rule, write the answer in terms of t alone (substitute $x(t)$, $y(t)$), or evaluate at the given t. ESA ★★ S003·19 · T003b 00:02:03
4. At a given t, a term vanishes if its dx/dt or dy/dt is 0 there (e.g. $dy/dt = \cos t = 0$ at $t = \pi/2$). ESA ★★ T003b 00:02:34

Why is it dz/dt (ordinary) and not $\partial z/\partial t$?
After substituting $x(t)$ and $y(t)$, z depends on the single variable t, so its derivative with respect to t is an ordinary derivative. The partial signs belong only to z with respect to x and y, where the other variable is held fixed.

S003·5 · T003a 00:02:23

Exam trap. Slide 003·6 has wrong numbers. For $z = x^2y^3$, $x = t^2$, $y = 2t^3$: $z_x = 2xy^3 = 16t^{11}$ (not $2t^{11}$), $z_y = 3x^2y^2 = 12t^{10}$ (not $9t^{10}$), and $dz/dt = 16t^{11}(2t) + 12t^{10}(6t^2) = 104t^{12}$ (not $60t^{12}$). Check: $z = 8t^{13}$. ADDED FOR CLARITY S003·6

Worked examples

W2.1 · Total derivative, both ways (slide 6, corrected) ESA ★★

GIVEN $z = x^2y^3$, $x = t^2$, $y = 2t^3$.

STEP 1 $z_x = 2xy^3 = 2t^2(8t^9) = 16t^{11}$, $z_y = 3x^2y^2 = 3t^4(4t^6) = 12t^{10}$.

STEP 2 $\dfrac{dx}{dt} = 2t$, $\dfrac{dy}{dt} = 6t^2$.

STEP 3 $\dfrac{dz}{dt} = 16t^{11}(2t) + 12t^{10}(6t^2) = 32t^{12} + 72t^{12}$.

ANSWER $\boxed{\dfrac{dz}{dt} = 104\,t^{12}}$

CHECK Substituting first: $z = t^4 \cdot 8t^9 = 8t^{13}$, so $dz/dt = 104t^{12}$. The slide's $60t^{12}$ is wrong.

ADDED FOR CLARITY S003·6

W2.2 · The lecture's version ESA ★

GIVEN $z = x^2y^3$, $x = t$, $y = 2t^2$.

STEP 1 Direct: $z = t^2(2t^2)^3 = 8t^8$, so $dz/dt = 64t^7$.

STEP 2 Rule: $z_x\,x' + z_y\,y' = 2xy^3(1) + 3x^2y^2(4t) = 16t^7 + 48t^7$.

ANSWER $\boxed{\dfrac{dz}{dt} = 64\,t^7}$

CHECK Both routes agree.

TRANSCRIPT ONLY T003a 00:02:05 · T003a 00:02:57

W2.3 · Total derivative at a point

ESA ★★

GIVEN $u = x^2 y\, e^{xy}$, $x = t^2 + 1$, $y = \sin t$; find du/dt at $t = \pi/2$.

STEP 1 $u_x = y\, e^{xy}(2x + x^2 y)$, $u_y = x^2 e^{xy}(1 + xy)$; $\dfrac{dx}{dt} = 2t$, $\dfrac{dy}{dt} = \cos t$.

STEP 2 At $t = \pi/2$: $x = \frac{\pi^2}{4} + 1$, $y = 1$, $dx/dt = \pi$, $dy/dt = 0$, so the second term vanishes.

STEP 3 $\dfrac{du}{dt} = \pi\, e^{x}(2x + x^2)$ with $x = \frac{\pi^2}{4} + 1$.

ANSWER $\boxed{\pi\, e^{\frac{\pi^2}{4}+1}\left[2\left(\frac{\pi^2}{4}+1\right) + \left(\frac{\pi^2}{4}+1\right)^2\right]}$

CHECK Direct differentiation in sympy agrees.

S003·18 · S003·19 · T003b 00:00:23

W2.4 · Total derivative in terms of t

ESA ★★

GIVEN $u = e^{xy} + \ln(y^2 + 1)$, $x = \tan t$, $y = \cos t$.

STEP 1 $u_x = y\, e^{xy}$, $u_y = x\, e^{xy} + \dfrac{2y}{y^2+1}$; $\dfrac{dx}{dt} = \sec^2 t$, $\dfrac{dy}{dt} = -\sin t$.

STEP 2 $\dfrac{du}{dt} = y\, e^{xy}\sec^2 t - \left(x\, e^{xy} + \dfrac{2y}{y^2+1}\right)\sin t$.

STEP 3 Substitute: $xy = \tan t \cos t = \sin t$ and $\cos t \sec^2 t - \tan t \sin t = \cos t$.

ANSWER $\boxed{\dfrac{du}{dt} = e^{\tan t \cos t}(\cos t \sec^2 t - \tan t \sin t) - \dfrac{2\cos t \sin t}{\cos^2 t + 1} = e^{\sin t}\cos t - \dfrac{\sin 2t}{1 + \cos^2 t}}$

CHECK The simplified form (added for clarity) equals the derivative of $e^{\sin t} + \ln(\cos^2 t + 1)$.

S003·20 · T003b 00:03:16

W2.5 · Total derivative at t = 0

ESA ★★

GIVEN $u = x^2 - y^2$, $x = e^t \cos t$, $y = e^t \sin t$; du/dt at $t = 0$.

STEP 1 $u_x = 2x$, $u_y = -2y$; $x' = e^t(\cos t - \sin t)$, $y' = e^t(\sin t + \cos t)$.

STEP 2 At $t = 0$: $x = 1$, $y = 0$, $x' = 1$, $y' = 1$.

STEP 3 $\dfrac{du}{dt} = 2(1)(1) - 2(0)(1)$.

ANSWER $\boxed{\left.\dfrac{du}{dt}\right|_{t=0} = 2}$

CHECK CW answer: 2.

CW U1 P10

W2.6 · Three intermediate variables

ESA ★★

GIVEN $f = x^3 + xz^2 + y^3 + xyz$, $x = e^t$, $y = \cos t$, $z = t^3$; df/dt at $t = 0$.

STEP 1 At $t = 0$: $x = 1$, $y = 1$, $z = 0$; $x' = 1$, $y' = 0$, $z' = 0$.

STEP 2 $f_x = 3x^2 + z^2 + yz = 3$ (the y and z terms drop out because $y' = z' = 0$).

STEP 3 $\dfrac{df}{dt} = f_x\,x' + f_y\,y' + f_z\,z' = 3(1) + 0 + 0.$

ANSWER $\boxed{\left.\dfrac{df}{dt}\right|_{t=0} = 3}$

CHECK CW answer: 3.

CW U1 P11

SHORT-NOTE RECAP

If $z = f(x, y)$ and $x = x(t)$, $y = y(t)$, then z is a function of t alone and its total derivative is $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$. A tree diagram with one branch per intermediate variable gives the terms. The answer is written in terms of t or evaluated at the given t.

3 The total differential: approximation and sensitivity

Without dividing by dt, the same rule gives the total differential $dz = f_x\,dx + f_y\,dy$. It approximates the change in z for small changes dx, dy, so it estimates function values near a known point. Comparing its coefficients shows which variable the output is most sensitive to. It also explains substitution in integrals.

Key terms and definitions

Total differential. For $z = f(x, y)$: $dz = f_x\,dx + f_y\,dy$. For $z = f(x_1, \ldots, x_n)$: $dz = f_{x_1}dx_1 + \cdots + f_{x_n}dx_n$. The total derivative is also called the differential. S004·3 · S004·4 · T004a 00:02:04

> *"The total derivative is also called the differential. The differential is used in integration by substitution and in approximation. It is also useful to calculate the Errors in approximation."* S004·4

Formulas

E3.1 · Total differential ESA ★★

$$dz = f_x(x, y)\,dx + f_y(x, y)\,dy$$

gives	Approximate change in z for small changes dx, dy.
applies	$z = f(x, y)$ differentiable.
note	n variables: $dz = f_{x_1}dx_1 + f_{x_2}dx_2 + \cdots + f_{x_n}dx_n$.

S004·3 · S004·4 · S003·8 · T003a 00:13:56

E3.2 · Approximating a nearby value ESA ★★

$$f(x_0 + dx,\, y_0 + dy) \approx f(x_0, y_0) + f_x(x_0, y_0)\,dx + f_y(x_0, y_0)\,dy$$

gives	Estimated function value near a known point.
applies	dx, dy small.
note	Slide 003·8: $f(2.1, -3.03) \approx 6 + 1.3(0.1) + (-0.6)(-0.03) = 6.148$.

S003·8 · T003a 00:13:09

Graphs and diagrams

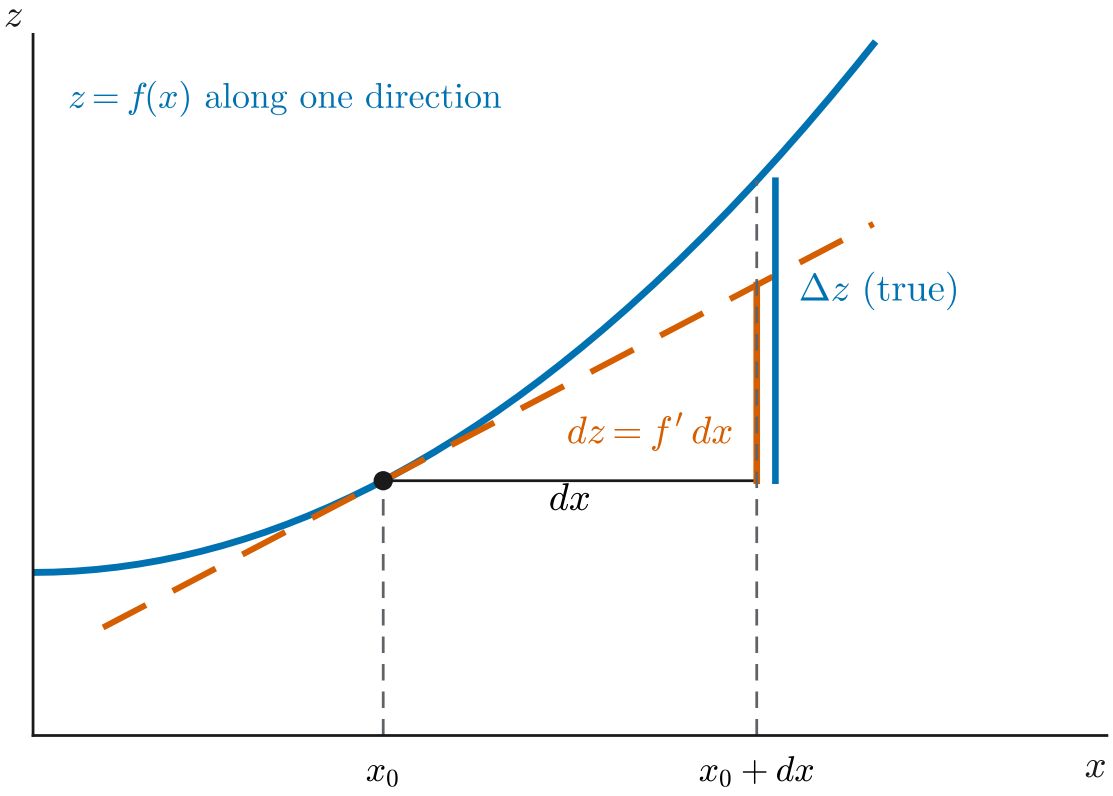
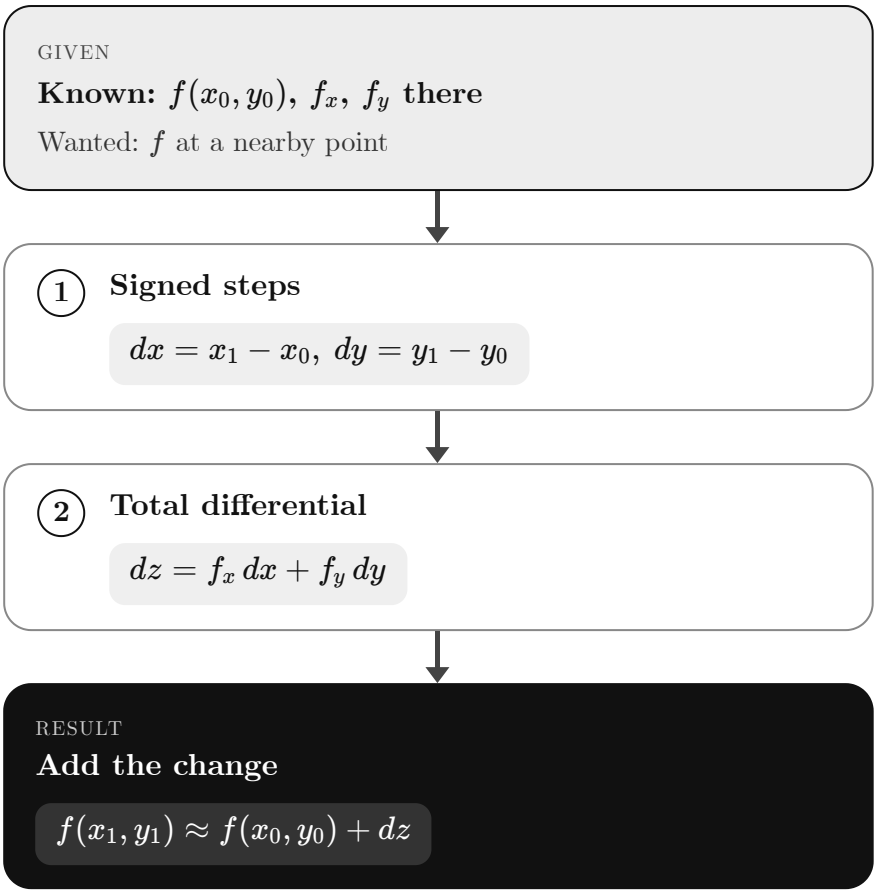


Figure G3.1. The differential approximates the true change. ESA ★★

ADDED FOR CLARITY S003·8 · S004·4

How to read it. Cut along one direction. The true change Δz follows the blue curve; the differential dz follows the dashed tangent line. For a small step dx the two are nearly equal, which is why $f(x_0 + dx) \approx f(x_0) + dz$. The gap grows as the step grows.

Flowcharts



Remark. Slide 003·8: $dz = 1.3(0.1) + (-0.6)(-0.03) = 0.148$, so $f(2.1, -3.03) \approx 6.148$.

Figure F3.1. Estimating a value near a known point. ESA ★★

S003·8 · T003a 00:13:09

Important points and why

1. The total differential estimates how much f changes from a known point to a nearby one. ESA ★★ S003·8
2. Sensitivity: in $dV = 40\pi\,dr + 4\pi\,dh$ the coefficient of dr is ten times that of dh, so the tank's volume is more sensitive to the radius than to the height. ESA ★★ S003·11 · S003·12 · T003a 00:18:30
3. Sometimes you need not finish the problem: the coefficients of the differential already tell you which variable contributes most. ESA ★ T003a 00:19:05
4. Integration by substitution uses the differential: with $t = x^2$, $dt = 2x\,dx$, so $\int e^{x^2} x\,dx = \frac{1}{2}\int e^t\,dt$. ESA ★ TRANSCRIPT ONLY T004a 00:07:57 · T004a 00:08:29
5. Errors and approximations, another application of the differential, come at the end of the unit. ESA ★ TRANSCRIPT ONLY T004a 00:08:47

Why does the tank's volume depend more on the radius?
$V = \pi r^2 h$ has $V_r = 2\pi r h = 40\pi$ and $V_h = \pi r^2 = 4\pi$ at $r = 2$, $h = 10$. So $dV = 4\pi(10\,dr + dh)$: a change in r is multiplied by ten times as much as the same change in h.

S003·11 · T003a 00:18:45

Exam trap. Use signed changes: from -3 to -3.03, $dy = -0.03$ (slide 003·8 writes 0.03 but correctly uses -0.03). S003·8

Worked examples

W3.1 · Approximate a function value

ESA ★★

given $f(2,-3) = 6$, $f_x(2,-3) = 1.3$, $f_y(2,-3) = -0.6$; approximate $f(2.1,-3.03)$.

step 1 $dx = 2.1 - 2 = 0.1$, $dy = -3.03 - (-3) = -0.03$.

step 2 $dz = 1.3(0.1) + (-0.6)(-0.03) = 0.13 + 0.018 = 0.148$.

step 3 $f(2.1,-3.03) \approx 6 + 0.148$.

answer $\boxed{f(2.1,-3.03) \approx 6.148}$

check Signs: $dy < 0$ and $f_y < 0$, so that term adds.

S003·8 · T003a 00:13:09

W3.2 · Which dimension matters more?

ESA ★★

given Cylindrical tank, $h = 10$ ft, diameter 4 ft ($r = 2$).

step 1 $V = \pi r^2 h$, $V_r = 2\pi r h$, $V_h = \pi r^2$.

step 2 $dV = 2\pi r h\,dr + \pi r^2\,dh = 40\pi\,dr + 4\pi\,dh$.

step 3 $40\pi \approx 125.7$ against $4\pi \approx 12.57$.

answer More sensitive to the radius (diameter) than to the height

check A diameter change of dD is $dr = dD/2$: still $20\pi\,dD$ against $4\pi\,dh$.

S003·11 · S003·12

W3.3 · Total differentials

ESA ★★

GIVEN (i) $u = xy$, (ii) $u = x/y$.

STEP 1 (i) $u_x = y$, $u_y = x$.

STEP 2 (ii) $u_x = 1/y$, $u_y = -x/y^2$.

ANSWER (i) $du = y\,dx + x\,dy$; (ii) $du = \dfrac{y\,dx - x\,dy}{y^2}$

CHECK Matches the CW key; (ii) is the quotient rule.

CW U1 P9

SHORT-NOTE RECAP

The total differential $dz = f_x\,dx + f_y\,dy$ (or $\sum f_{x_i} dx_i$) gives the approximate change in z for small changes in the variables. It approximates nearby values, $f(x_0 + dx, y_0 + dy) \approx f + f_x\,dx + f_y\,dy$. It also measures sensitivity: the larger coefficient marks the variable that matters most. It is used in substitution and error estimates.

4 Rates of change problems

Many word problems ask how fast one quantity changes when others change at known rates. Write the quantity as a function of the changing variables, take the total derivative with respect to time, and substitute the given values. “Remains constant” means its derivative is 0. A constraint linking the variables is differentiated implicitly.

Key terms and definitions

Rate of change. The derivative of a quantity with respect to time t; positive means increasing, negative means decreasing. T003a 00:12:06

Formulas

E4.1 · Constant quantity ESA ★★

$$A = \text{constant} \implies \frac{dA}{dt} = 0$$

gives	The equation that fixes the unknown rate.
applies	“Remains constant” in the question.

S003·9 · T003a 00:16:32

Graphs and diagrams

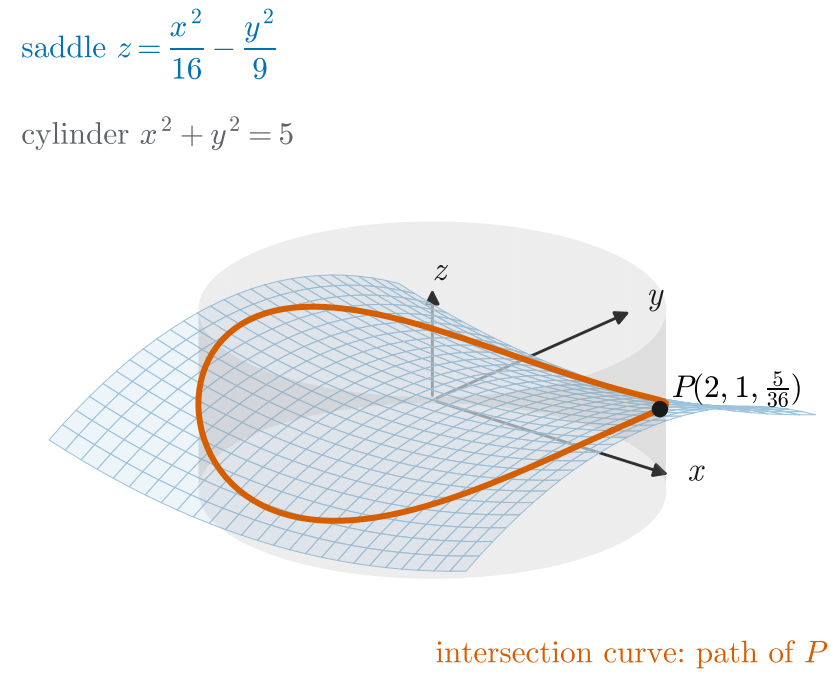


Figure G4.1. A point moving on a saddle and a cylinder. ESA ★★

S003·7 · T003a 00:06:27

How to read it. The saddle $z = x^2/16 - y^2/9$ (light blue) and the cylinder $x^2 + y^2 = 5$ (grey) meet in the vermilion curve. The point P moves along that curve; at $x = 2$, $y = 1$ its height rises at $dz/dt = 5/36$ cm/s when $dx/dt = 0.2$ cm/s (W4.1). The slide calls the surface a paraboloid, but it is a hyperbolic paraboloid (a saddle).

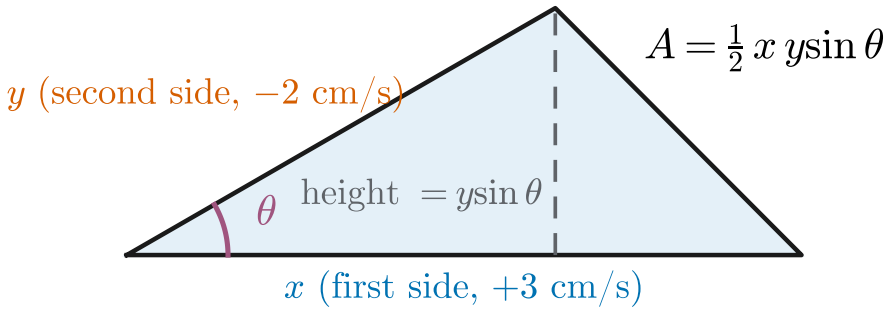
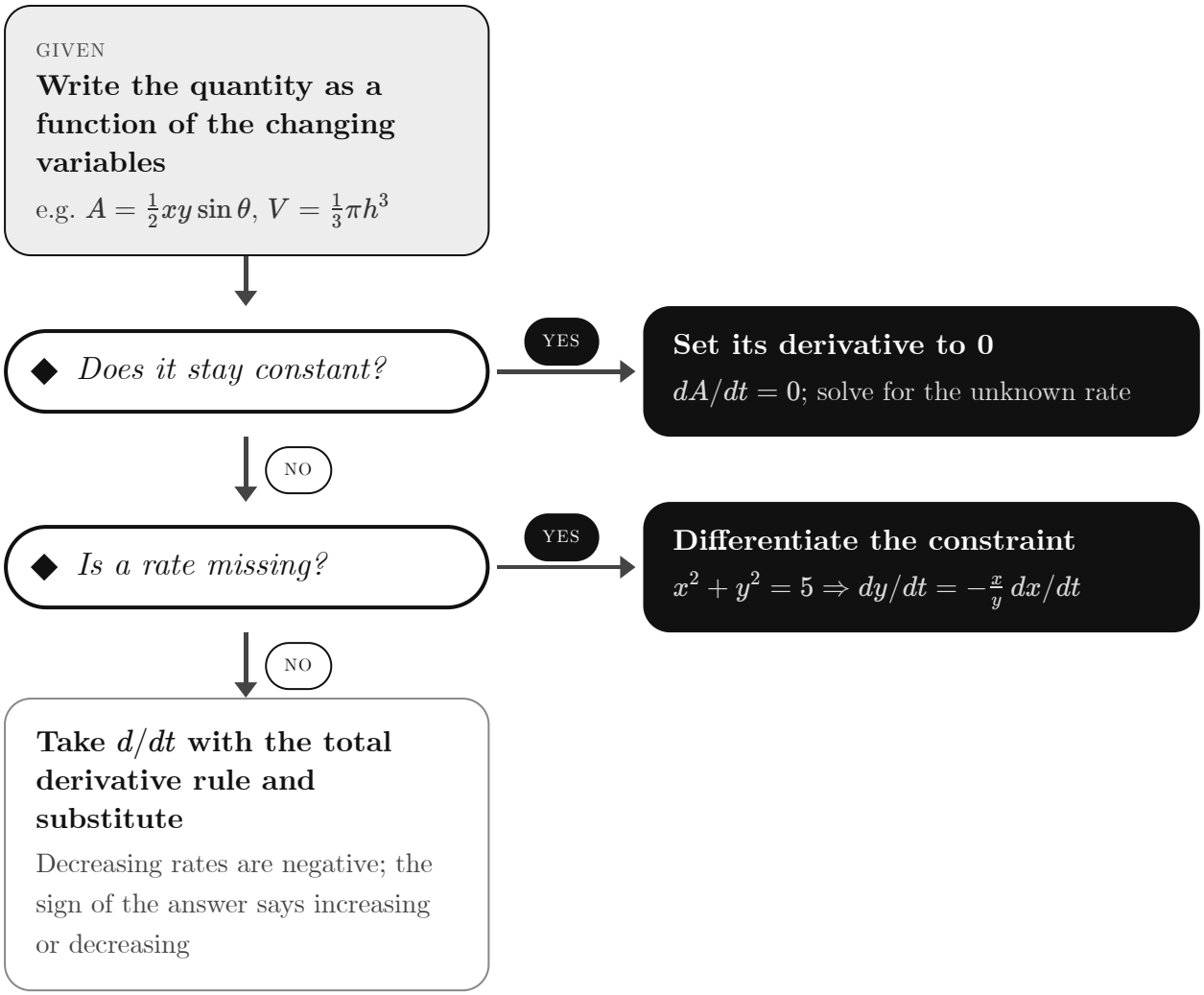


Figure G4.2. Triangle with two sides and the included angle. ESA ★★

S003·9 · T003a 00:15:25

How to read it. Sides x and y meet at the angle θ. The dashed perpendicular is the height $y \sin\theta$, so the area is $A = \frac{1}{2} x y \sin\theta$. Label the three changing quantities x, y, θ before differentiating.

Flowcharts



Remark. Both checks can apply in one problem; work down the ladder, then substitute the numbers last.

Figure F4.1. Solving a rates-of-change problem. ESA ★★

S003·7 · S003·9 · S003·21 · T003a 00:07:25

Important points and why

1. If $dz/dt > 0$ the quantity is increasing at that instant; if $dz/dt < 0$ it is decreasing. ESA ★★ T003a 00:12:06
2. When an unknown rate such as dy/dt is needed, find it from the constraint by implicit differentiation, e.g. $x^2 + y^2 = 5 \Rightarrow dy/dt = -\frac{x}{y}\,dx/dt$. ESA ★★ S003·7 · T003a 00:09:29
3. A decreasing side has a negative rate: "decreasing at 2 cm/s" means $dy/dt = -2$. ESA ★★ S003·10

Why does the angle have to decrease in the triangle problem?
$A = \frac{1}{2}xy\sin\theta$. One side grows ($+3$) and the other shrinks (-2), but at the given instant their combined effect $\frac{1}{2}\sin\theta(3y - 2x) = \frac{1}{2}\cdot\frac{1}{2}(90 - 40) > 0$ would increase the area. To keep A constant, θ must decrease.

ADDED FOR CLARITY S003·9 · S003·10

Exam trap. Slide 003·10 prints $d\theta/dt = 0.04831$. The value is $-25/(300\sqrt{3}) = -\sqrt{3}/36 \approx -0.0481$ rad/s: negative, and 481, not 4831. ADDED FOR CLARITY S003·10 · T003a 00:16:55

Exam trap. $x^2/16 - y^2/9 = z$ is a hyperbolic paraboloid (a saddle), not a cup-shaped paraboloid. The calculation does not change. ADDED FOR CLARITY S003·7 · T003a 00:06:27

Worked examples

W4.1 · Point moving on a saddle and a cylinder — ESA ★★

GIVEN $z = \dfrac{x^2}{16} - \dfrac{y^2}{9}$, $x^2 + y^2 = 5$, $\dfrac{dx}{dt} = 0.2$ cm/s; find dz/dt at $x = 2$ (so $y = 1$).

STEP 1 $\dfrac{dz}{dt} = \dfrac{x}{8}\dfrac{dx}{dt} - \dfrac{2y}{9}\dfrac{dy}{dt}$.

STEP 2 Constraint: $2x + 2y\,y' = 0$, so $\dfrac{dy}{dt} = -\dfrac{x}{y}\dfrac{dx}{dt} = -0.4$.

STEP 3 $\dfrac{dz}{dt} = \dfrac{x}{8}(0.2) + \dfrac{2x}{9}(0.2)$: the y cancels. At $x = 2$: $0.05 + 0.0889$.

ANSWER $\boxed{\dfrac{dz}{dt} = \dfrac{5}{36} \approx 0.139 \text{ cm/s (increasing)}}$

CHECK Positive, so z is increasing at $x = 2$.

S003·7 · T003a 00:06:27 · T003a 00:11:32

W4.2 · Angle of a triangle of constant area — ESA ★★

GIVEN $x = 20$, $y = 30$, $\theta = \pi/6$; $\dfrac{dx}{dt} = 3$, $\dfrac{dy}{dt} = -2$; area constant.

STEP 1 $A = \frac{1}{2}xy\sin\theta$ (height $= y\sin\theta$).

STEP 2 $0 = \dfrac{dA}{dt} = \frac{1}{2}\Big(y\sin\theta(3) + x\sin\theta(-2) + xy\cos\theta\,\dfrac{d\theta}{dt}\Big)$.

STEP 3 $= \frac{1}{2}\big(45 - 20 + 300\sqrt{3}\,\theta'\big)$, so $\theta' = -\dfrac{25}{300\sqrt{3}}$.

ANSWER $\boxed{\dfrac{d\theta}{dt} = -\dfrac{\sqrt{3}}{36} \approx -0.0481 \text{ rad/s (decreasing)}}$

CHECK The slide's "0.04831" is a misprint (see Exam trap).

S003·9 · S003·10 · T003a 00:14:31

W4.3 · Growing triangle

ESA ★★

GIVEN $b = 10$, $h = 6$, $db/dt = 2$, $dh/dt = 1$ (cm, cm/s).

STEP 1 $A = \frac{1}{2}bh$, $\dfrac{dA}{dt} = \frac{1}{2}\left(b\dfrac{dh}{dt} + h\dfrac{db}{dt}\right)$.

STEP 2 $= \frac{1}{2}(10 \cdot 1 + 6 \cdot 2) = \frac{1}{2}(22)$.

ANSWER $\boxed{\dfrac{dA}{dt} = 11 \text{ cm}^2/\text{s}}$

CHECK Units: cm × cm/s = cm²/s.

S003·21 · S003·22 · T003b 00:04:57

W4.4 · Filling a cone

ESA ★★

GIVEN $dV/dt = 9$ cm³/s, $r = h$ always, $h = 6$ cm.

STEP 1 $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi h^3$.

STEP 2 $\dfrac{dV}{dt} = \pi h^2 \dfrac{dh}{dt}$, so $9 = 36\pi\, \dfrac{dh}{dt}$.

ANSWER $\boxed{\dfrac{dh}{dt} = \dfrac{1}{4\pi} \approx 0.0796 \text{ cm/s}}$

CHECK Use $r = h$ before differentiating.

S003·23 · S003·24 · T003b 00:06:41

W4.5 · Growing rectangle

ESA ★★

GIVEN Sides 5 ft and 4 ft, increasing at 1.5 ft/s and 0.5 ft/s.

STEP 1 $A = xy$, $\dfrac{dA}{dt} = y\dfrac{dx}{dt} + x\dfrac{dy}{dt} = 4(1.5) + 5(0.5)$.

ANSWER $\boxed{\dfrac{dA}{dt} = 8.5 \text{ ft}^2/\text{s}}$

CHECK CW answer: 8.5 sq. ft/s.

CW U1 P12

W4.6 · Changing cone

ESA ★★

GIVEN $h = 15$ cm, $dh/dt = 0.2$; $r = 10$ cm, $dr/dt = -0.3$.

STEP 1 $V = \frac{1}{3}\pi r^2 h$, $\dfrac{dV}{dt} = \frac{1}{3}\pi\left(2rh\dfrac{dr}{dt} + r^2\dfrac{dh}{dt}\right)$.

STEP 2 $= \frac{1}{3}\pi(2 \cdot 10 \cdot 15 \cdot (-0.3) + 100 \cdot 0.2) = \frac{1}{3}\pi(-90 + 20)$.

ANSWER $\boxed{\dfrac{dV}{dt} = -\dfrac{70\pi}{3}\ \text{cm}^3/\text{s (decreasing)}}$

CHECK CW answer: $-70\pi/3$.

CW U1 P13

SHORT-NOTE RECAP

For a rate problem, write the quantity as a function of the changing variables, take d/dt of it with the total derivative rule, and substitute the given values and rates (decreasing means negative). If something stays constant its derivative is 0; a constraint between variables gives a missing rate by implicit differentiation.

5 The chain rule for composite functions

If $z = f(x, y)$ and x, y are functions of two variables u, v, then z is a composite function of u and v. Its partial derivatives follow every path in the tree diagram: $z_u = z_x x_u + z_y y_u$ and $z_v = z_x x_v + z_y y_v$. When f is unknown, f_x, f_y stay as symbols. Substitution tricks ($r = 2x - 3y$, ...) and polar coordinates are the standard exam forms.

Key terms and definitions

Chain rule (several variables). If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables u, v, then $\dfrac{\partial z}{\partial u} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}$ and $\dfrac{\partial z}{\partial v} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial v} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial v}$. S003·13 · T003a 00:21:07

The chain of a problem. The list of dependencies, e.g. "u depends on x, y; x, y depend on r, t; so u is a composite function of r, t". Knowing it fixes which chain-rule formula to write. T004b 00:02:31

> *"We can define chain rule for a function of two or more independent variable: If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables say u, v then we can find the partial derivative of z with respect to u and v."* S003·13

Formulas

E5.1 · Chain rule, two intermediate variables ESA ★★★

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

gives	Partial derivatives of a composite function.
applies	$z = f(x, y)$, $x = g(u, v)$, $y = h(u, v)$.
note	All derivatives are partial, because everything depends on two variables.

S003·13 · S003·14 · T003a 00:21:07

E5.2 · Chain rule, three intermediate variables ESA ★★★

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial v}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial v}{\partial t}\frac{\partial t}{\partial x}$$

gives	One term per intermediate variable.
applies	$v = f(r, s, t)$, each of r, s, t a function of x, y, z.
note	A term is 0 when that intermediate variable does not contain x.

S003·15 · T003a 00:23:10

E5.3 · Function of one intermediate variable ESA ★★

$$u = f(r),\ r = \sqrt{x^2+y^2+z^2} \implies \frac{\partial u}{\partial x} = f'(r)\,\frac{\partial r}{\partial x} = f'(r)\,\frac{x}{r}$$

GIVES $\partial u/\partial x$ in terms of $f'(r)$.

APPLIES u depends on x, y, z only through r.

S004·12 · T004b 00:05:07

E5.4 · Polar coordinates: derivatives you reuse ESA ★★

$$\begin{aligned} &x = r\cos\theta,\ y = r\sin\theta: && x_\theta = -r\sin\theta, \quad y_\theta = r\cos\theta \\ &r = \sqrt{x^2+y^2},\ \theta = \tan^{-1}\tfrac{y}{x}: && r_x = \frac{x}{r},\ r_y = \frac{y}{r},\ \theta_x = -\frac{y}{r^2},\ \theta_y = \frac{x}{r^2} \end{aligned}$$

GIVES The inner derivatives for polar problems.

APPLIES $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}(y/x)$.

NOTE So $u_x = f_r\frac{x}{r} - f_\theta\frac{y}{r^2}$ and $u_y = f_r\frac{y}{r} + f_\theta\frac{x}{r^2}$ for $u = f(r,\theta)$.

S004·13 · S004·15 · S004·17

E5.5 · Polar identity ESA ★★

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2$$

GIVES The squared gradient in polar coordinates.

APPLIES $u = u(x,y)$, $x = r\cos\theta$, $y = r\sin\theta$.

NOTE Proved in W5.16.

CW U1 P15

Graphs and diagrams

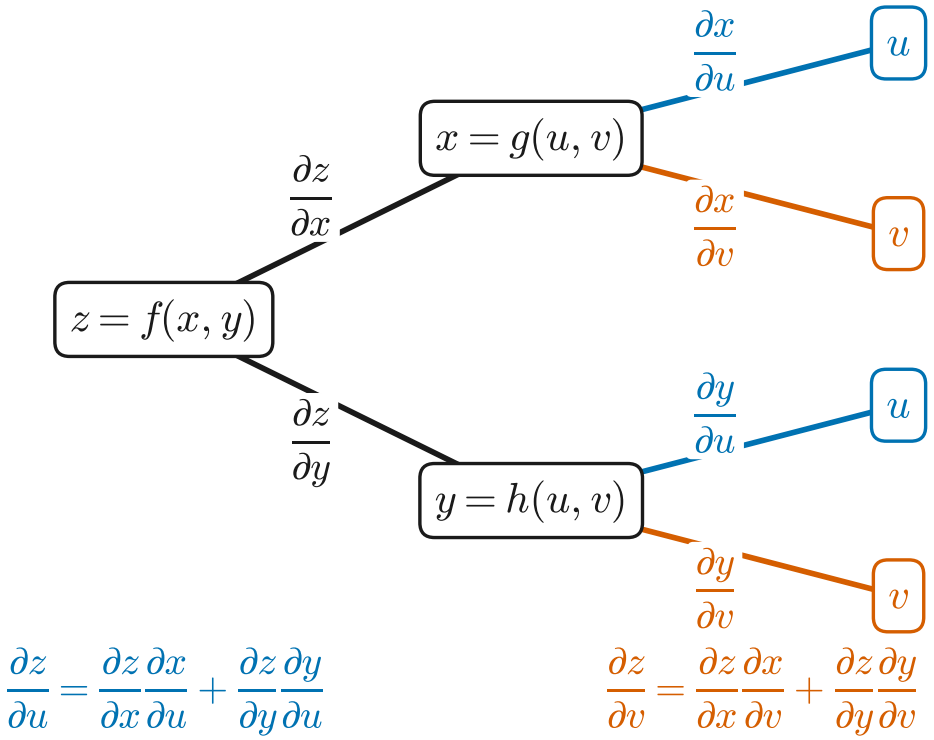
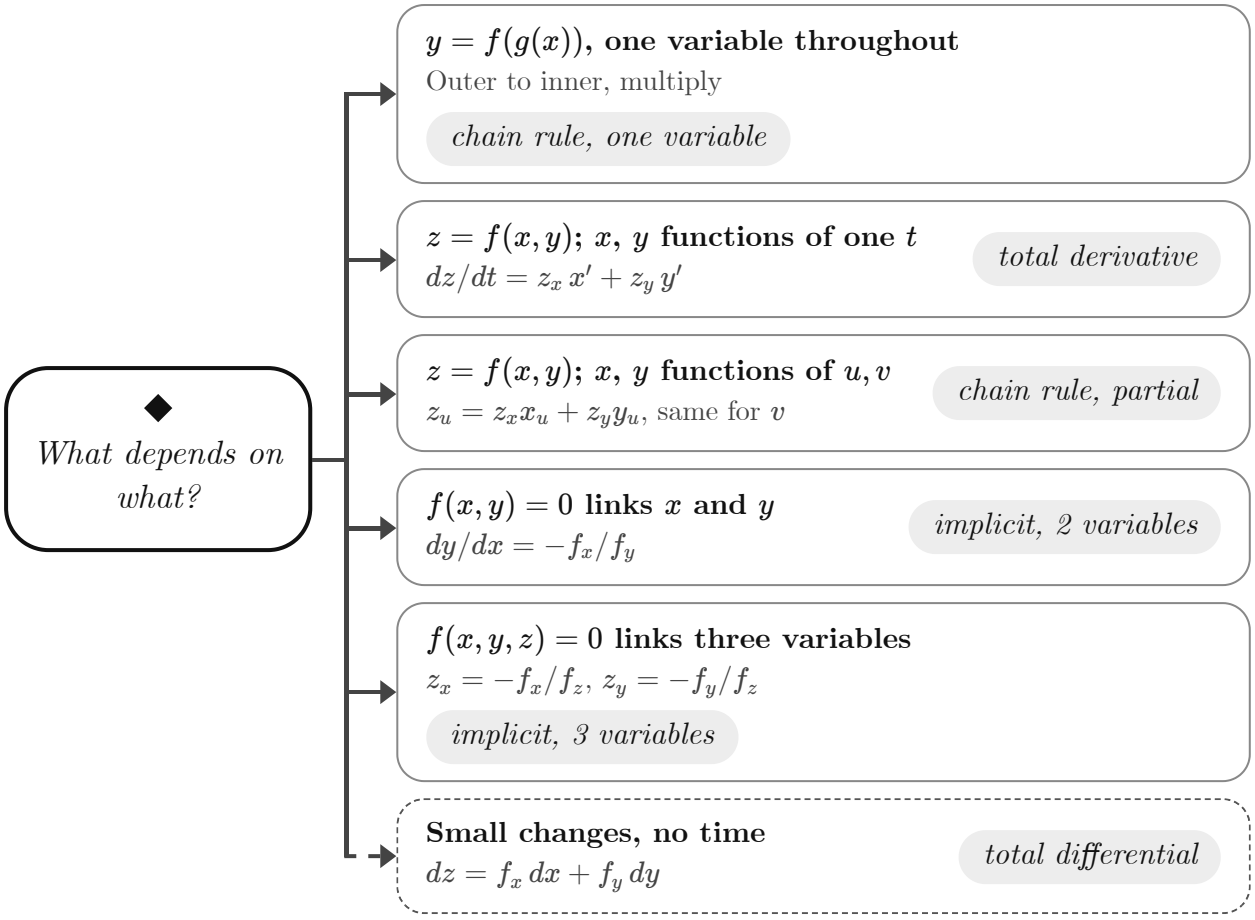


Figure G5.1. Tree diagram for the chain rule. ESA ★★★

S003·14 · T003a 00:19:42

How to read it. z branches to x and y, and each of those branches to both u and v, giving four leaves. For $\partial z/\partial u$ follow only the two paths ending at u (blue): $z_x x_u + z_y y_u$. For $\partial z/\partial v$ follow the two ending at v (vermilion). In an exam, draw the tree first and read the terms off it.

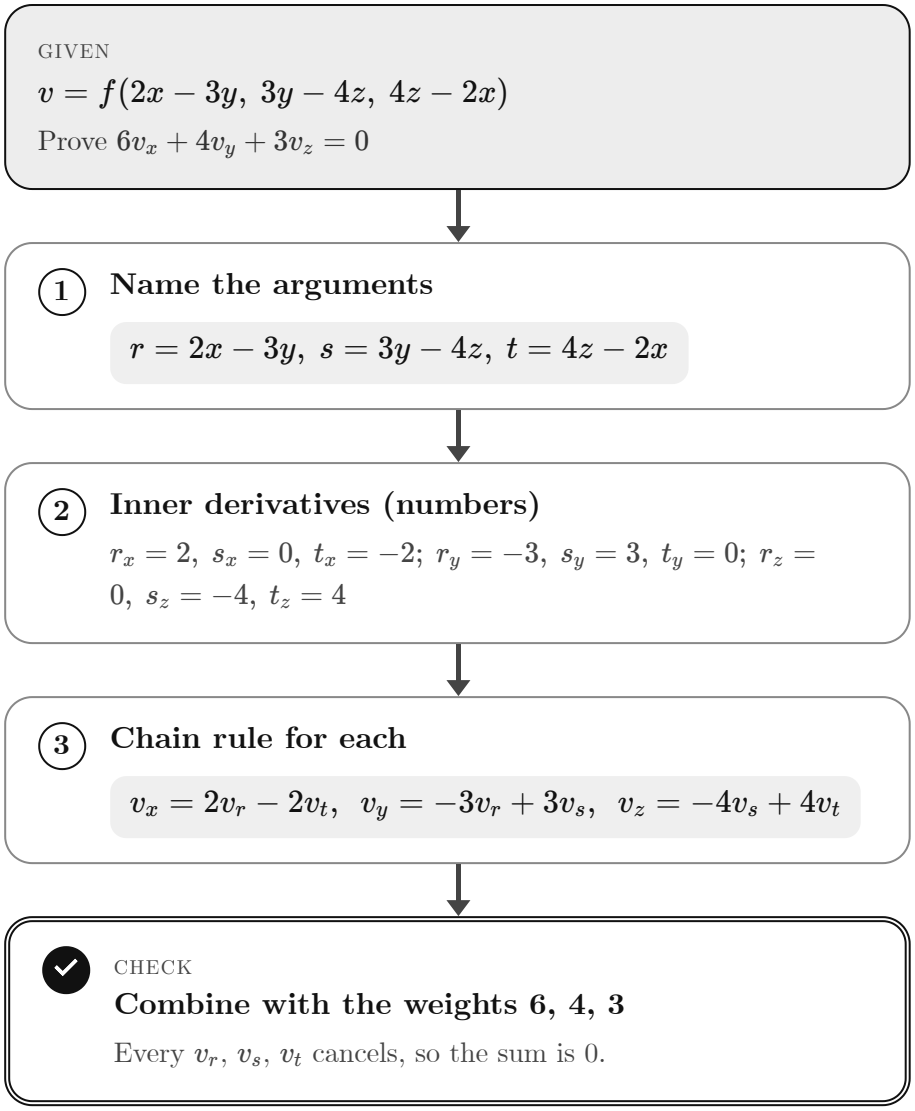
Flowcharts



Remark. The dashed route has no derivative with respect to a variable: it estimates a change. Everything else starts from the tree diagram.

Figure F5.1. Which rule? Read the chain first. ESA ★★★

S003·5 · S003·13 · S004·5 · S004·6 · T004b 00:02:31



Exam trap. Slide 003·16 drops the 3 in $v_y = -3v_r + 3v_s$; without it the terms do not cancel.

Figure F5.2. Proving an identity such as $6v_x + 4v_y + 3v_z = 0$. ESA ★★★

S003·15 · S003·16 · ESA Jan-24 s2 1d

Total derivative vs chain rule vs implicit differentiation

	Total derivative	Chain rule (partial)	Implicit function
Set-up	$z = f(x, y)$, $x(t)$, $y(t)$	$z = f(x, y)$, $x(u, v)$, $y(u, v)$	$f(x, y) = 0$ or $f(x, y, z) = 0$
Final variables	one: t	two: u, v	x (and y)
Formula	$z_x\, x' + z_y\, y'$	$z_x x_u + z_y y_u$	$-f_x/f_y$, $-f_x/f_z$
Result written	dz/dt	$\partial z/\partial u$, $\partial z/\partial v$	dy/dx, $\partial z/\partial x$
Typical question	rate of change at an instant	prove an identity	slope or tangent line

ADDED FOR CLARITY S003·5 · S003·13 · S004·4 · S004·6

Important points and why

1. When x and y depend on two variables u, v, z cannot be differentiated "totally": only z_u and z_v exist. ESA ★★ T003a 00:20:40
2. Write the tree diagram every time a function has more than one independent variable, so that no partial derivative is missed. ESA ★★★ T003a 00:25:02
3. It is essential to know the chain given in the problem: what depends on what. The formula follows from it. ESA ★★★ T004b 00:02:31
4. If f is not given ($u = f(x, y)$), keep f_x, f_y (or f_r, f_θ) as symbols in the answer. ESA ★★★ S003·17 · S003·25 · T003a 00:26:28
5. For $f(2x - 3y,\, 3y - 4z,\, 4z - 2x)$, name the arguments r, s, t and compute $r_x, s_x, t_x, \ldots$; most are constants or 0. ESA ★★★ S003·15 · T003a 00:22:33
6. When u is a composite function of r, θ (or s, t), eliminate x and y from the final answer: you must know why you substitute. ESA ★★★ T004b 00:07:52 · T004b 00:10:28
7. Functions of $x^2 + y^2$ alone (e.g. $\sin(x^2 + y^2)$, $e^{x^2+y^2}$) have $\partial u/\partial\theta = 0$ in polar coordinates, since $x^2 + y^2 = r^2$ does not involve θ. ESA ★★ ADDED FOR CLARITY S004·10 · S004·13 · S004·14
8. $\tan^{-1}(y/x)$ is exactly θ in polar coordinates, so $\partial u/\partial\theta = 1$. ESA ★★ ADDED FOR CLARITY S004·15
9. The chain rule extends to any number of variables: z of x, y, each of x, y of three variables, and so on. ESA ★ T003a 00:21:40

Why does $u = f(x - y,\, y - z,\, z - x)$ give $u_x + u_y + u_z = 0$?
Each argument is a difference whose derivatives sum to zero: for $r = x - y$, $r_x + r_y + r_z = 1 - 1 + 0 = 0$. Adding the three chain-rule expressions gives $f_r(0) + f_s(0) + f_t(0)$. In words: moving x, y, z by the same amount changes none of the arguments.

ADDED FOR CLARITY ESA Jan-24 s2 1d

Why do the chain-rule answers keep f_r and f_θ?
The outer function f is unknown, so its partial derivatives cannot be computed; only the inner derivatives ($r_s = t$, $\theta_s = s/\sqrt{s^2+t^2}$, ...) can.

S003·17 · T003a 00:26:28

Exam trap. Slide 003·16 shows $\dfrac{\partial v}{\partial y} = -3\dfrac{\partial v}{\partial r} + \dfrac{\partial v}{\partial s}$; it should be $-3\dfrac{\partial v}{\partial r} + 3\dfrac{\partial v}{\partial s}$ because $s = 3y - 4z$ gives $s_y = 3$. With the 3 the proof works. ADDED FOR CLARITY S003·16

Exam trap. Do not stop with x and y still in an answer that should be in r, θ or s, t. T004b 00:07:52

Exam trap. Write $\dfrac{\partial z}{\partial u}$, not $\dfrac{dz}{du}$: z depends on both u and v. ADDED FOR CLARITY S003·13

Worked examples

W5.1 · Prove $6v_x + 4v_y + 3v_z = 0$ ESA ★★★

GIVEN $v = f(2x - 3y,\ 3y - 4z,\ 4z - 2x)$.

STEP 1 Let $r = 2x - 3y$, $s = 3y - 4z$, $t = 4z - 2x$: $r_x = 2,\ s_x = 0,\ t_x = -2$; $r_y = -3,\ s_y = 3,\ t_y = 0$; $r_z = 0,\ s_z = -4,\ t_z = 4$.

STEP 2 $v_x = 2v_r - 2v_t$, $v_y = -3v_r + 3v_s$, $v_z = -4v_s + 4v_t$.

STEP 3 $6v_x + 4v_y + 3v_z = (12 - 12)v_r + (12 - 12)v_s + (-12 + 12)v_t$.

ANSWER $\boxed{6v_x + 4v_y + 3v_z = 0}$

CHECK The weights 6, 4, 3 are chosen so that each of v_r, v_s, v_t cancels.

S003·15 · S003·16 · T003a 00:22:14

W5.2 · Unknown outer function ESA ★★★

GIVEN $z = f(r, \theta)$, $r = ts - t^2$, $\theta = \sqrt{s^2 + t^2}$.

STEP 1 $r_s = t$, $\theta_s = \dfrac{s}{\sqrt{s^2+t^2}}$; $r_t = s - 2t$, $\theta_t = \dfrac{t}{\sqrt{s^2+t^2}}$.

STEP 2 $z_s = f_r\, r_s + f_\theta\, \theta_s$ and $z_t = f_r\, r_t + f_\theta\, \theta_t$.

ANSWER $\boxed{\dfrac{\partial z}{\partial s} = t\, f_r + \dfrac{s}{\sqrt{s^2+t^2}} f_\theta, \quad \dfrac{\partial z}{\partial t} = (s - 2t) f_r + \dfrac{t}{\sqrt{s^2+t^2}} f_\theta}$

CHECK sympy's chain rule on a symbolic f gives the same.

S003·17 · T003a 00:25:02

W5.3 · $u = f(x,y)$, $x = e^{st}$, $y = s^2 + t^2$ ESA ★★★

GIVEN $u = f(x,y)$, $x = e^{st}$, $y = s^2 + t^2$.

STEP 1 $x_s = te^{st}$, $x_t = se^{st}$, $y_s = 2s$, $y_t = 2t$.

STEP 2 $u_s = f_x x_s + f_y y_s$, $u_t = f_x x_t + f_y y_t$.

ANSWER $\boxed{\dfrac{\partial u}{\partial s} = te^{st} f_x + 2s\, f_y, \quad \dfrac{\partial u}{\partial t} = se^{st} f_x + 2t\, f_y}$

S003·25 · T003b 00:08:18

W5.4 · $\sin(x^2+y^2)$ in polar form ESA ★★

GIVEN $u = \sin(x^2+y^2)$, $x = r\cos t$, $y = r\sin t$; find u_t.

STEP 1 $u_x = 2x\cos(x^2+y^2)$, $u_y = 2y\cos(x^2+y^2)$; $x_t = -r\sin t$, $y_t = r\cos t$.

STEP 2 $u_t = 2\cos(r^2)\,[\,r\cos t(-r\sin t) + r\sin t(r\cos t)\,]$.

ANSWER $\boxed{\dfrac{\partial u}{\partial t} = 0}$

CHECK $u = \sin r^2$ does not involve t at all.

S004·10 · T004b 00:00:10

W5.5 · Three intermediate variables ESA ★★

GIVEN $u = x^2y + yz^3$, $x = r\cos\theta$, $y = r\sin\theta$, $z = \theta$; find u_θ.

STEP 1 $u_x = 2xy$, $u_y = x^2 + z^3$, $u_z = 3yz^2$; $x_\theta = -r\sin\theta$, $y_\theta = r\cos\theta$, $z_\theta = 1$.

STEP 2 $u_\theta = 2xy(-r\sin\theta) + (x^2+z^3)(r\cos\theta) + 3yz^2$.

STEP 3 Substitute: $-2r^3\cos\theta\sin^2\theta + r^3\cos^3\theta = r^3\cos\theta(3\cos^2\theta - 2)$.

ANSWER $\boxed{\dfrac{\partial u}{\partial \theta} = r^3\cos\theta\,(3\cos^2\theta - 2) + r\,(3\theta^2\sin\theta + \theta^3\cos\theta)}$

CHECK sympy agrees.

S004·11 · T004b 00:03:09

W5.6 · $u = f(r)$ with r the distance

ESA ★★

GIVEN $u = f(r)$, $r = \sqrt{x^2+y^2+z^2}$.

STEP 1 $u_x = \dfrac{du}{dr}\, r_x = f'(r)\, r_x$.

STEP 2 $r_x = \dfrac{x}{\sqrt{x^2+y^2+z^2}} = \dfrac{x}{r}$.

ANSWER $\boxed{\dfrac{\partial u}{\partial x} = f'(r)\,\dfrac{x}{r}}$

CHECK By symmetry $u_y = f'(r)\,y/r$, $u_z = f'(r)\,z/r$.

S004·12 · T004b 00:05:07

W5.7 · $e^{x^2+y^2}$ in polar form

ESA ★★

GIVEN $u = e^{x^2+y^2}$, $x = r\cos\theta$, $y = r\sin\theta$.

STEP 1 $u_\theta = 2xe^{x^2+y^2}(-r\sin\theta) + 2ye^{x^2+y^2}(r\cos\theta)$.

STEP 2 $= 2re^{r^2}(-r\cos\theta\sin\theta + r\sin\theta\cos\theta)$.

ANSWER $\boxed{\dfrac{\partial u}{\partial \theta} = 0}$

CHECK $u = e^{r^2}$ has no θ.

S004·13 · S004·14 · T004b 00:06:38

W5.8 · Show $\partial u/\partial\theta = 1$

ESA ★★

GIVEN $u = \tan^{-1}(y/x)$, $x = r\cos\theta$, $y = r\sin\theta$.

STEP 1 $u_x = \dfrac{-y}{x^2+y^2}$, $u_y = \dfrac{x}{x^2+y^2}$; $x_\theta = -r\sin\theta$, $y_\theta = r\cos\theta$.

STEP 2 $u_\theta = \dfrac{r^2\sin^2\theta + r^2\cos^2\theta}{r^2}$.

ANSWER $\boxed{\dfrac{\partial u}{\partial \theta} = 1}$

CHECK $u = \tan^{-1}(\tan\theta) = \theta$.

S004·15 · T004b 00:08:40

W5.9 · $u = \sin(xy) + y^3,\ x = e^s,\ y = s + t$ ESA ★★

GIVEN $u = \sin(xy) + y^3,\ x = e^s,\ y = s + t.$

STEP 1 $u_x = y\cos(xy),\ u_y = x\cos(xy) + 3y^2;\ x_s = e^s,\ y_s = 1,\ x_t = 0,\ y_t = 1.$

STEP 2 $u_s = y\cos(xy)e^s + (x\cos(xy) + 3y^2),\ u_t = x\cos(xy) + 3y^2,$ then substitute $x = e^s,\ y = s + t.$

ANSWER $\boxed{u_s = (s+t)e^s\cos\big(e^s(s+t)\big) + e^s\cos\big(e^s(s+t)\big) + 3(s+t)^2,\quad u_t = e^s\cos\big(e^s(s+t)\big) + 3(s+t)^2}$

CHECK sympy agrees.

S004·16 · T004b 00:09:32

W5.10 · $u = f(r,\theta)$ from Cartesian coordinates ESA ★★

GIVEN $u = f(r,\theta),\ r = \sqrt{x^2+y^2},\ \theta = \tan^{-1}(y/x).$

STEP 1 $r_x = \dfrac{x}{r},\ r_y = \dfrac{y}{r},\ \theta_x = -\dfrac{y}{r^2},\ \theta_y = \dfrac{x}{r^2}.$

STEP 2 $u_x = f_r r_x + f_\theta \theta_x,\ u_y = f_r r_y + f_\theta \theta_y.$

ANSWER $\boxed{\dfrac{\partial u}{\partial x} = f_r\dfrac{x}{r} - f_\theta\dfrac{y}{r^2},\quad \dfrac{\partial u}{\partial y} = f_r\dfrac{y}{r} + f_\theta\dfrac{x}{r^2}}$

S004·17 · T004b 00:10:47

W5.11 · $xu_x + yu_y + zu_z$ for $f(x/y,\ y/z,\ z/x)$ ESA ★★★

GIVEN $u = f\big(\tfrac{x}{y}, \tfrac{y}{z}, \tfrac{z}{x}\big);$ find $xu_x + yu_y + zu_z.$

STEP 1 Let $p = \tfrac{x}{y},\ q = \tfrac{y}{z},\ w = \tfrac{z}{x}.$

STEP 2 $xu_x = x\big(\tfrac{1}{y}f_p - \tfrac{z}{x^2}f_w\big) = p\,f_p - w\,f_w.$

STEP 3 $yu_y = y\big(-\tfrac{x}{y^2}f_p + \tfrac{1}{z}f_q\big) = -p\,f_p + q\,f_q.$

STEP 4 $zu_z = z\big(-\tfrac{y}{z^2}f_q + \tfrac{1}{x}f_w\big) = -q\,f_q + w\,f_w.$

STEP 5 Add the three lines.

ANSWER $\boxed{x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = 0}$

CHECK u is homogeneous of degree 0 (scaling x, y, z leaves every ratio unchanged), so Euler's theorem gives 0 too.

ADDED FOR CLARITY ESA Jan-24 s1 1d

W5.12 · $u_x + u_y + u_z$ for $f(x-y,\, y-z,\, z-x)$ ESA ★★★

GIVEN $u = f(x-y,\ y-z,\ z-x)$; find $u_x + u_y + u_z$.

STEP 1 Let $p = x-y$, $q = y-z$, $w = z-x$.

STEP 2 $u_x = f_p - f_w$, $u_y = -f_p + f_q$, $u_z = -f_q + f_w$.

STEP 3 Add.

ANSWER $\boxed{\dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y} + \dfrac{\partial u}{\partial z} = 0}$

CHECK Same pattern as W5.1 with equal weights.

ADDED FOR CLARITY ESA Jan-24 s2 1d

W5.13 · $z = f(u,v)$, $u = x^2 - y^2$, $v = 2xy$ ESA ★★★

GIVEN Show $z_x^2 + z_y^2 = 4(u^2+v^2)^{1/2}\,(z_u^2 + z_v^2)$.

STEP 1 $z_x = z_u(2x) + z_v(2y)$, $z_y = z_u(-2y) + z_v(2x)$.

STEP 2 $z_x^2 + z_y^2 = 4(x^2+y^2)(z_u^2+z_v^2)$: the cross terms $8xy\, z_u z_v$ cancel.

STEP 3 $u^2 + v^2 = (x^2-y^2)^2 + 4x^2y^2 = (x^2+y^2)^2$, so $x^2+y^2 = (u^2+v^2)^{1/2}$.

ANSWER $\boxed{z_x^2 + z_y^2 = 4\sqrt{u^2+v^2}\,(z_u^2+z_v^2)}$

CHECK Numerical check with a sample f in sympy: residual $\approx 10^{-14}$.

ADDED FOR CLARITY ESA Jan-24 s3 1b

W5.14 · $z = f(x,y)$, $x = e^u \sin v$, $y = e^u \cos v$ ESA ★★★

GIVEN Show $z_x^2 + z_y^2 = e^{-2u}(z_u^2 + z_v^2)$.

STEP 1 $z_u = z_x\, e^u \sin v + z_y\, e^u \cos v = x z_x + y z_y$.

STEP 2 $z_v = z_x\, e^u \cos v - z_y\, e^u \sin v = y z_x - x z_y$.

STEP 3 $z_u^2 + z_v^2 = (x^2+y^2)(z_x^2+z_y^2)$: the cross terms cancel.

STEP 4 $x^2 + y^2 = e^{2u}$.

ANSWER $\boxed{z_x^2 + z_y^2 = e^{-2u}\,(z_u^2+z_v^2)}$

CHECK sympy with a sample f: exact 0.

ADDED FOR CLARITY ESA Jan-24 s4 1b

W5.15 · $z = \log(u^2 + v)$ ESA ★★★

GIVEN $z = \log(u^2+v)$, $u = e^{x+y^2}$, $v = x + y^2$; find $2y\,z_x - z_y$.

STEP 1 $z_u = \dfrac{2u}{u^2+v}$, $z_v = \dfrac{1}{u^2+v}$; $u_x = u$, $u_y = 2yu$, $v_x = 1$, $v_y = 2y$.

STEP 2 $z_x = \dfrac{2u^2+1}{u^2+v}$, $z_y = \dfrac{2y(2u^2+1)}{u^2+v}$.

ANSWER $\boxed{2y\,\dfrac{\partial z}{\partial x} - \dfrac{\partial z}{\partial y} = 0}$

CHECK CW answer: 0. Both depend on x, y only through $x + y^2$.

ESA Jun-24 1a · CW U1 P14

W5.16 · Polar identity ESA ★★

GIVEN $u = u(x,y)$, $x = r\cos\theta$, $y = r\sin\theta$.

STEP 1 $u_r = u_x\cos\theta + u_y\sin\theta$.

STEP 2 $u_\theta = -u_x\, r\sin\theta + u_y\, r\cos\theta$, so $\frac{1}{r}u_\theta = -u_x\sin\theta + u_y\cos\theta$.

STEP 3 Square and add: the cross terms cancel and $\sin^2\theta + \cos^2\theta = 1$.

ANSWER $\boxed{u_x^2 + u_y^2 = u_r^2 + \dfrac{1}{r^2}\,u_\theta^2}$

CHECK sympy with a sample u: residual $\approx 10^{-16}$.

CW U1 P15

Past-paper questions

ESA Jan 2024 (5 marks) UE23MA141A set 1, Q1.d · solved in W5.11

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$.

ESA Jan 2024 (5 marks) UE23MA141A set 2, Q1.d · solved in W5.12

If $u = f(x-y,\, y-z,\, z-x)$, find the value of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$.

ESA Jan 2024 (6 marks) UE23MA141A set 3, Q1.b · solved in W5.13

If $z = f(u,v)$ and $u = x^2 - y^2$, $v = 2xy$, show that $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 4(u^2+v^2)^{1/2}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

ESA Jan 2024 (6 marks) UE23MA141A set 4, Q1.b · solved in W5.14

If $z = f(x,y)$ and $x = e^u\sin v$, $y = e^u\cos v$, show that $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{-2u}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

ESA June 2024 | 6 marks | UE23MA141A Q1.a · solved in W5.15

If $z = \log(u^2 + v)$, where $u = e^{x+y^2}$, $v = x + y^2$, then find the value of $2y\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}$.

SHORT-NOTE RECAP

If $z = f(x, y)$ with $x = g(u, v)$, $y = h(u, v)$, then z is a composite function of u, v and $z_u = z_x x_u + z_y y_u$, $z_v = z_x x_v + z_y y_v$ (one term per path in the tree diagram). With three intermediate variables there are three terms. If f is unknown, keep f_x, f_y as symbols. Write the final answer in the new variables. Standard results: $f(x - y, y - z, z - x)$ gives $u_x + u_y + u_z = 0$; in polar form $u_x^2 + u_y^2 = u_r^2 + u_\theta^2/r^2$.

6 Implicit functions of two variables

A relation $f(x,y)=c$ defines y as a function of x without solving for it; geometrically it is a level curve. Its total differential is zero, $f_x\,dx+f_y\,dy=0$, which gives $dy/dx=-f_x/f_y$ with no product or quotient rule. The slope then gives tangent lines, and the same idea handles awkward relations such as $(\cos x)^y=(\sin y)^x$.

Key terms and definitions

Implicit function. A function given by a relation $f(x,y)=c$ (for example $x^3y+3xy^2=3$) rather than by a formula $y=g(x)$. S004·3 · T004a 00:00:18

Level surface (level curve). The set where $z=f(x,y)=c$ (or $f(x_1,\dots,x_n)=c$). Examples: equipotential surfaces, isotherms. The cylinders $x^2+y^2=c$ and the spheres $x^2+y^2+z^2=c$ are layers, one for each c. S004·4 · T004a 00:02:35 · T004a 00:03:29

> *"If z represents a level surface, i.e. if $z=f(x,y)=c$, then $dz=0$. The total derivative is also called the differential."* S004·4

> *"Setting $\dfrac{dz}{dx}=0$ leads to a simple computation formula for implicit differentiation: $f_x\cdot 1+f_y\dfrac{dy}{dx}=0$."* S004·5

Formulas

E6.1 · Implicit differentiation, two variables ESA ★★

$$f(x,y)=c \implies \frac{dy}{dx}=-\frac{f_x}{f_y}, \qquad f_y\neq 0$$

gives	Slope of the level curve at a point.
applies	$f_y\neq 0$ at the point.
note	From $df=f_x\,dx+f_y\,dy=0$. Slide 004·4 prints the fraction upside down.

S004·5 · S004·6 · T004a 00:06:36 · T003b 00:10:33

E6.2 · Tangent line at a point ESA ★★

$$y-y_0=m\,(x-x_0), \qquad m=\left.\frac{dy}{dx}\right|_{(x_0,y_0)}$$

gives	Equation of the tangent to the curve at (x_0,y_0).
applies	(x_0,y_0) on the curve.
note	Through the origin: $y=mx$.

S004·7 · T004a 00:14:25

E6.3 · Derivative of a constant power and a variable power ESA ★★

$$\frac{d}{dx}a^x = a^x \ln a, \qquad \frac{d}{dx}x^n = n\,x^{n-1}$$

GIVES	The two rules needed for $(\cos x)^y$ and $(\sin y)^x$.
APPLIES	In a partial derivative, the other variable is a constant.
NOTE	ln needs a positive argument: $\cos x > 0$, $\sin y > 0$.

S004·8 · T004a 00:16:43

Graphs and diagrams

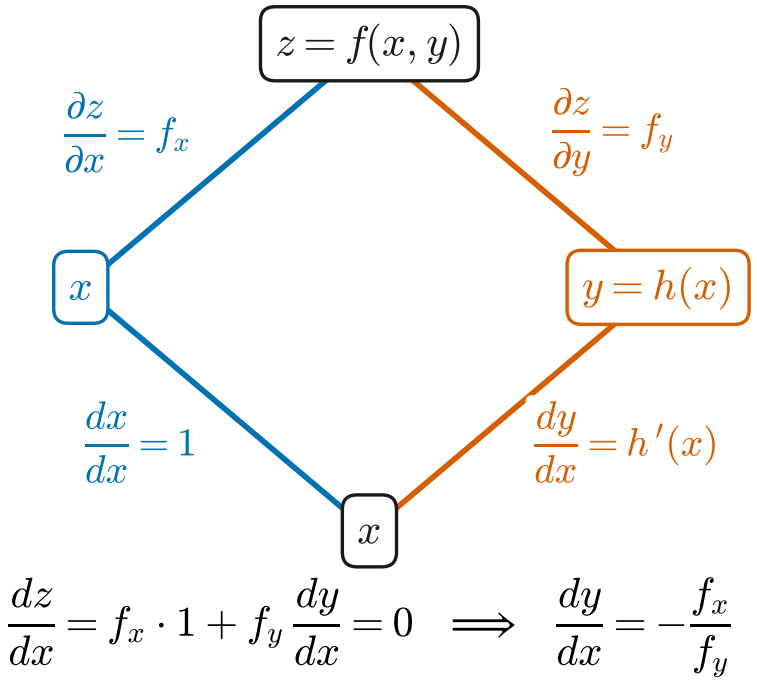


Figure G6.1. Branch diagram for implicit differentiation. ESA ★★

S004·5 · T004a 00:07:57

How to read it. Differentiate $z = f(x, y)$ with respect to x along both branches. The left branch is direct ($f_x \cdot dx/dx = f_x \cdot 1$). The right branch goes through $y = h(x)$ ($f_y \cdot dy/dx$). On a level curve the total is 0: $f_x + f_y\, dy/dx = 0$.

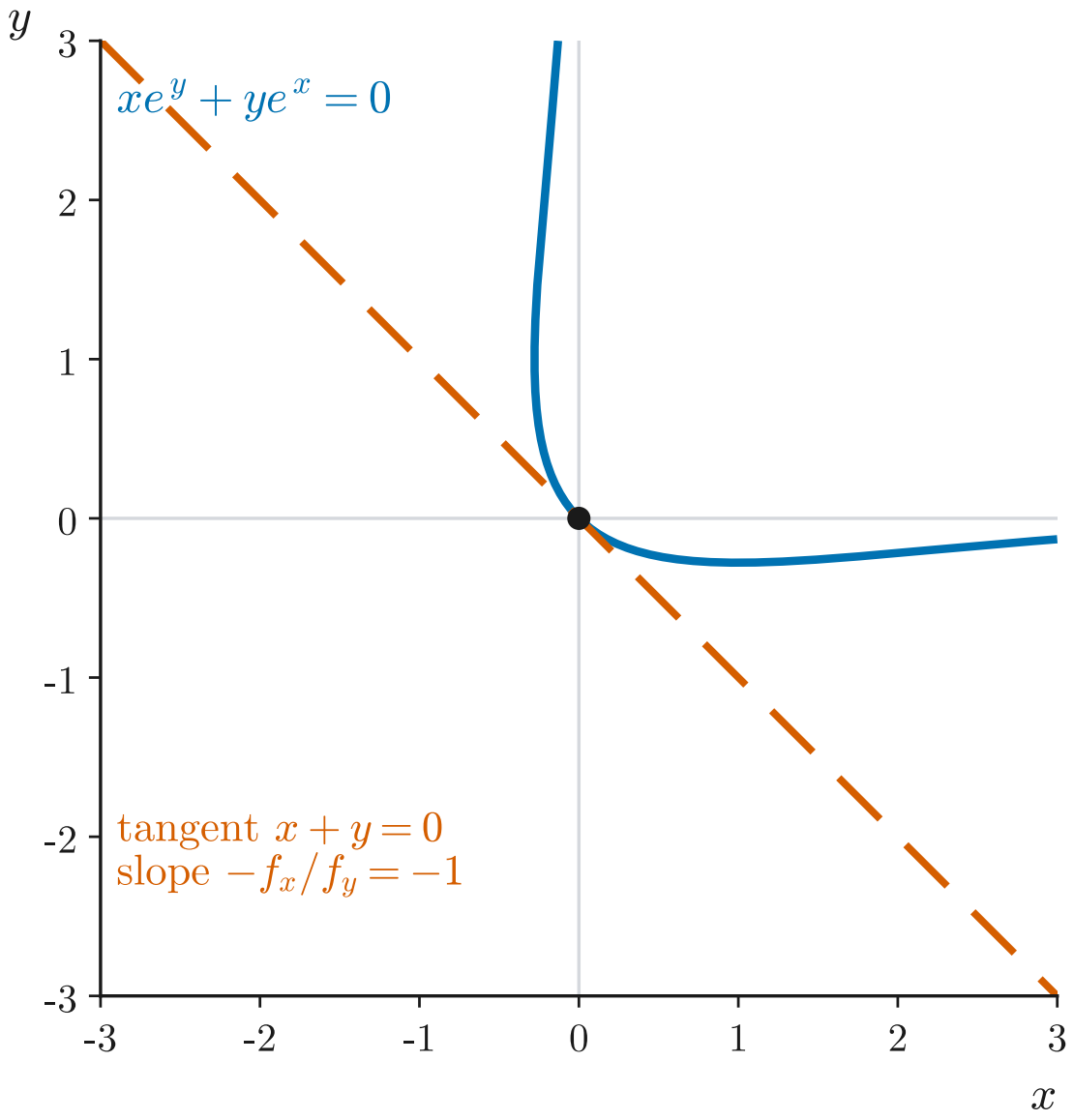


Figure G6.2. The curve $xe^y + ye^x = 0$ and its tangent at the origin. ESA ★★

S004·7 · T004a 00:13:48

How to read it. The blue curve is the set where $f(x, y) = xe^y + ye^x = 0$. At the origin $dy/dx = -f_x/f_y = -1$, so the dashed tangent is $y = -x$. The curve is plotted from its equation (the slide's picture is a 3-D grapher screenshot).

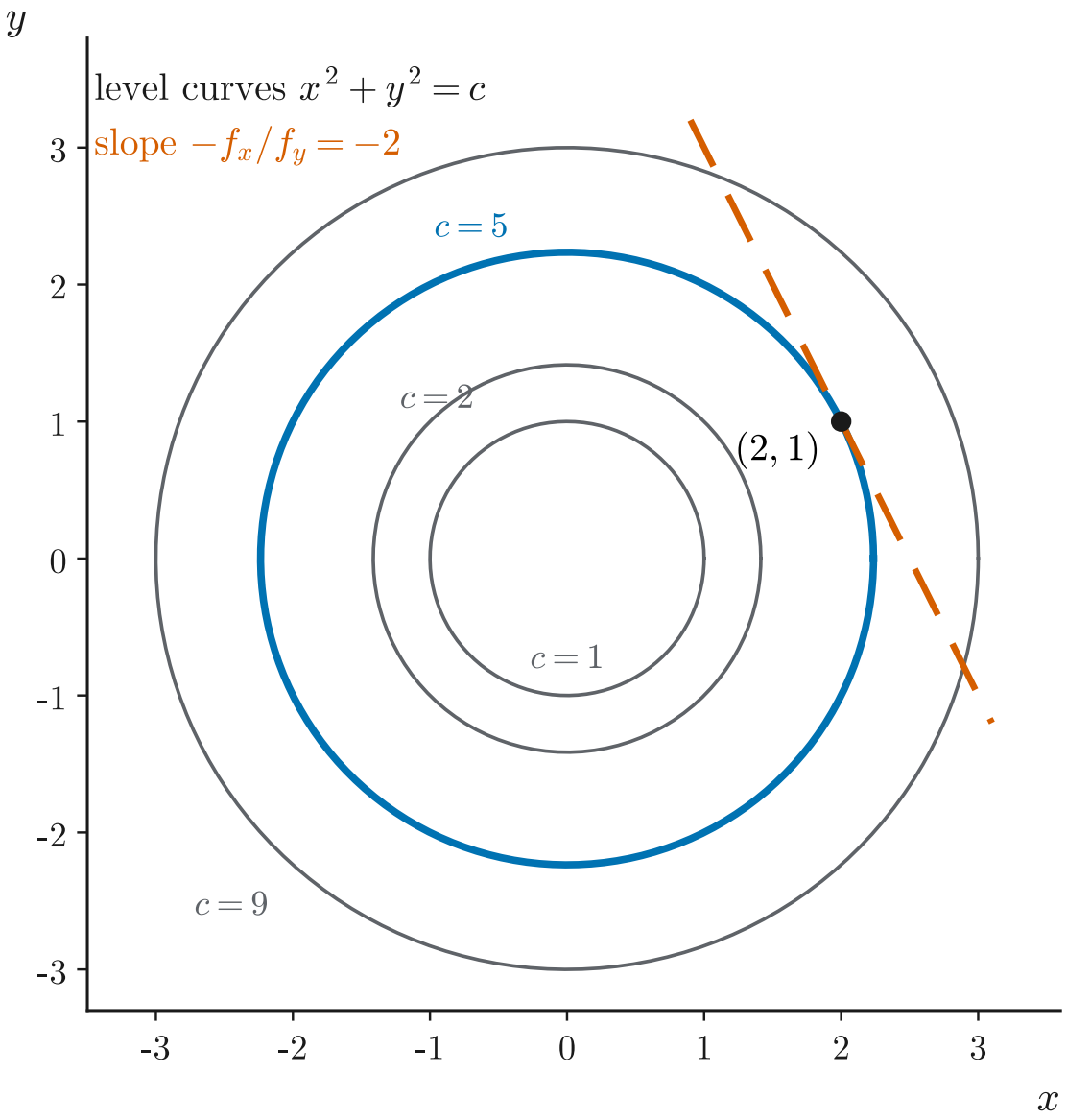
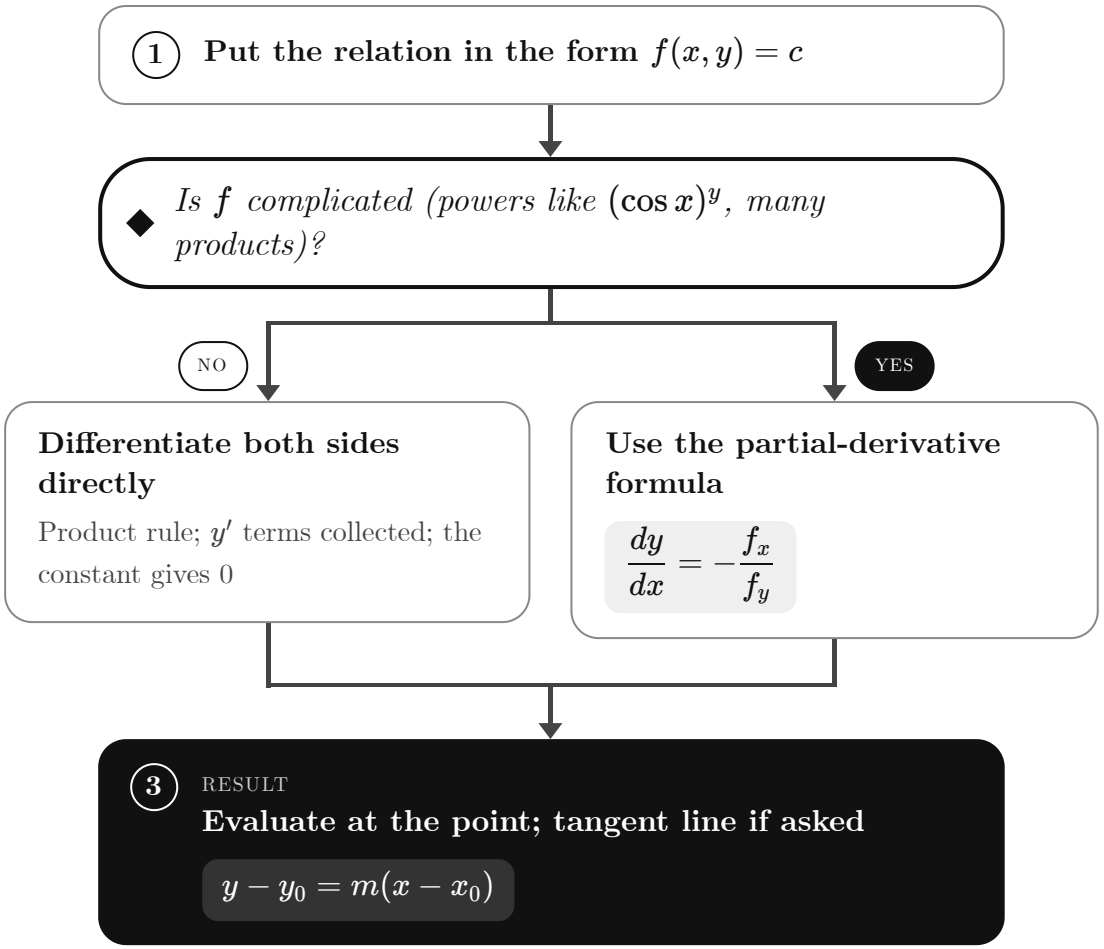


Figure G6.3. Level curves and the slope $-f_x/f_y$. ESA ★★

ADDED FOR CLARITY S004·4 · S003·7 · T004a 00:03:10

How to read it. Level curves of $f = x^2 + y^2$ for $c = 1, 2, 5, 9$: the cross-sections of the cylinders on slide 004·4. On $c = 5$ at $(2, 1)$ the dashed tangent has slope $-f_x/f_y = -2x/2y = -2$, the same $dy/dx = -x/y$ used in W4.1.

Flowcharts



Remark. Both routes give the same dy/dx (W6.1, W6.2). The formula needs $f_y \neq 0$ at the point.

Figure F6.1. Finding dy/dx for an implicit function: two routes. ESA ★★

S003·27 · S004·3 · S004·5 · T003b 00:10:33 · T004a 00:06:19

Important points and why

1. Differentiating an implicit relation directly needs the product rule on every term, and the right-hand constant gives 0. ESA ★★ S004·3 · T004a 00:00:48
2. The formula $dy/dx = -f_x/f_y$ gives the same answer and is far easier when f is complicated. ESA ★★ T004a 00:06:19 · T004a 00:06:36
3. Restriction to remember: the formula needs $z = f(x,y)$ to be a constant (a level curve). Move everything to one side first: $f(x,y)=c$ or $f(x,y)=0$. ESA ★★★ T004a 00:01:49 · T004a 00:02:04
4. There are two ways to solve "find dy/dx": differentiate both sides directly, or use $-f_x/f_y$. Both are accepted. ESA ★★ S003·27 · T003b 00:10:33
5. The slope dy/dx at a point gives the tangent line $y-y_0=m(x-x_0)$. ESA ★★ S004·7 · T004a 00:12:01
6. For $(\cos x)^y=(\sin y)^x$ the result needs $\cos x>0$ and $\sin y>0$, because of the logarithms. ESA ★ T004a 00:18:12

7. Implicit and composite together: if $u = \tan(x^2 + y^2)$ and $x^2 - y^2 = 2$, use the relation to eliminate y (or find dy/dx) before differentiating. ESA ★★ CW U1 P16 · CW U1 P18

Why is $dy/dx = -f_x/f_y$?
Along the curve $f(x, y) = c$ the value of f does not change, so $df = f_x\,dx + f_y\,dy = 0$. Divide by dx: $f_x + f_y\,\dfrac{dy}{dx} = 0$. The branch diagram (Figure G6.1) shows the same sum.
S004·4 · S004·5 · T004a 00:07:39

Why is the level-curve slope $-f_x/f_y$ and not $-f_y/f_x$ (as printed on slide 004·4)?
From $f_x + f_y\,y' = 0$ you get $y' = -f_x/f_y$. Test it on $x^2 + y^2 = 5$ at $(2, 1)$: $-f_x/f_y = -2x/(2y) = -2$, which matches differentiating directly; $-f_y/f_x$ would give $-1/2$.
ADDED FOR CLARITY S004·4 · S004·5

Exam trap. Slide 004·4 prints $dy/dx = -f_y(x, y)/f_x(x, y)$. The correct formula is $dy/dx = -f_x/f_y$. ADDED FOR CLARITY S004·4

Exam trap. In $(\cos x)^y$, with respect to x it is a power ($y(\cos x)^{y-1}(-\sin x)$); with respect to y it is an exponential ($(\cos x)^y \ln\cos x$). S004·8 · T004a 00:16:02

Worked examples

W6.1 · Direct differentiation vs the formula — ESA ★★

GIVEN $x^3y + 3xy^2 = 3$.

STEP 1 Direct: $x^3y' + 3x^2y + 3y^2 + 6xy\,y' = 0$.

STEP 2 Formula: $f_x = 3x^2y + 3y^2$, $f_y = x^3 + 6xy$.

STEP 3 $y' = -\dfrac{3x^2y + 3y^2}{x^3 + 6xy}$.

ANSWER $\boxed{\dfrac{dy}{dx} = -\dfrac{3y(x^2 + y)}{x(x^2 + 6y)}}$

CHECK Both methods agree; the slide's printed numerator is unclear.

S004·3 · T004a 00:00:48 · T004a 00:05:41

W6.2 · Two ways for $x^2 + xy + y^2 = 7$ ESA ★★

GIVEN $x^2 + xy + y^2 = 7$.

STEP 1 Direct: $2x + (xy' + y) + 2y\,y' = 0$, so $(x + 2y)y' = -(2x + y)$.

STEP 2 Formula: $f_x = 2x + y$, $f_y = x + 2y$, $y' = -f_x/f_y$.

ANSWER $\boxed{\dfrac{dy}{dx} = -\dfrac{2x + y}{x + 2y}}$

CHECK Same by both methods.

S003·27 · T003b 00:10:23

W6.3 · Tangent line at the origin ESA ★★

GIVEN $xe^y + ye^x = 0$; tangent at $(0, 0)$.

STEP 1 $f_x = e^y + ye^x$, $f_y = xe^y + e^x$.

STEP 2 $\dfrac{dy}{dx} = -\dfrac{e^y + ye^x}{xe^y + e^x}$; at $(0,0)$: $-\dfrac{1 + 0}{0 + 1} = -1$.

STEP 3 $y - 0 = -1(x - 0)$.

ANSWER $\boxed{x + y = 0}$

CHECK $(0, 0)$ satisfies the equation.

S004·7 · T004a 00:11:48

W6.4 · $(\cos x)^y = (\sin y)^x$ ESA ★★

GIVEN $f(x, y) = (\cos x)^y - (\sin y)^x = 0$.

STEP 1 $f_x = y(\cos x)^{y-1}(-\sin x) - (\sin y)^x \ln \sin y$.

STEP 2 $f_y = (\cos x)^y \ln \cos x - x(\sin y)^{x-1} \cos y$.

STEP 3 $y' = -f_x/f_y$. Divide through by $(\cos x)^y = (\sin y)^x$, then multiply top and bottom by $\cos x \sin y$.

ANSWER $\boxed{\dfrac{dy}{dx} = \dfrac{\sin y\,(y \sin x + \cos x \ln \sin y)}{\cos x\,(\sin y \ln \cos x - x \cos y)}}$

CHECK Checked numerically at a point on the curve ($x = 0.5$, $y \approx 0.909$): both give -0.4980.

S004·8 · T004a 00:14:57

W6.5 · Slope at (1, 1) ESA ★★

GIVEN $e^y - e^x + xy = 1$ at $(1, 1)$.

STEP 1 Check: $e - e + 1 = 1$, so the point is on the curve.

STEP 2 $f_x = -e^x + y$, $f_y = e^y + x$.

STEP 3 $y' = -\dfrac{-e+1}{e+1}$.

ANSWER $\boxed{\dfrac{dy}{dx} = \dfrac{e-1}{e+1}}$

CHECK CW answer: $(e-1)/(e+1)$.

CW U1 P17

W6.6 · Composite with an implicit constraint ESA ★★

GIVEN $u = \tan(x^2 + y^2)$ with $x^2 - y^2 = 2$; find du/dx.

STEP 1 From the relation, $y^2 = x^2 - 2$, so $u = \tan(2x^2 - 2)$.

STEP 2 $\dfrac{du}{dx} = \sec^2(2x^2 - 2) \cdot 4x$.

ANSWER $\boxed{\dfrac{du}{dx} = 4x \sec^2(2x^2 - 2)}$

CHECK Via the chain rule: $u_x + u_y\, y'$ with $y' = x/y$ gives the same. CW answer agrees.

CW U1 P16

W6.7 · Total derivative along an implicit curve ESA ★★

GIVEN $u = x \log(xy)$ with $x^3 + y^3 - 3xy - 1 = 0$; find du/dx.

STEP 1 $\dfrac{du}{dx} = u_x + u_y \dfrac{dy}{dx}$, with $u_x = \log(xy) + 1$, $u_y = \dfrac{x}{y}$.

STEP 2 From the relation: $\dfrac{dy}{dx} = -\dfrac{3x^2 - 3y}{3y^2 - 3x} = \dfrac{y - x^2}{y^2 - x}$.

ANSWER $\boxed{\dfrac{du}{dx} = 1 + \log(xy) + \dfrac{x\,(y - x^2)}{y\,(y^2 - x)}}$

CHECK CW answer agrees; sympy residual 0.

CW U1 P18

SHORT-NOTE RECAP

An implicit function is given by a relation $f(x,y)=c$, a level curve. Along it $df = f_x\,dx + f_y\,dy = 0$, so $\dfrac{dy}{dx} = -\dfrac{f_x}{f_y}$ $(f_y \neq 0)$. Direct differentiation of both sides gives the same result. The slope at (x_0, y_0) gives the tangent $y - y_0 = m(x - x_0)$; e.g. $xe^y + ye^x = 0$ has tangent $x + y = 0$ at the origin.

7 Implicit functions of three variables

A relation $f(x,y,z)=0$ is a level surface and makes z a function of x and y. Holding y fixed gives $\partial z/\partial x = -f_x/f_z$. Holding z fixed gives $\partial y/\partial x = -f_x/f_y$. Each needs the denominator to be non-zero, and the point used must lie on the surface.

Key terms and definitions

Implicit function of three variables. $f(x,y,z)=0$ with z (or y) regarded as a function of the remaining variables; the other variables are intermediate variables. S004·6 · T004a 00:09:05

> *"Let $f(x,y,z)=0$ be the equation of an implicit function of three variables x,y,z. Suppose y and z are functions of x, then f is a function of one independent variable x and y, z are intermediate variables."* S004·6

Formulas

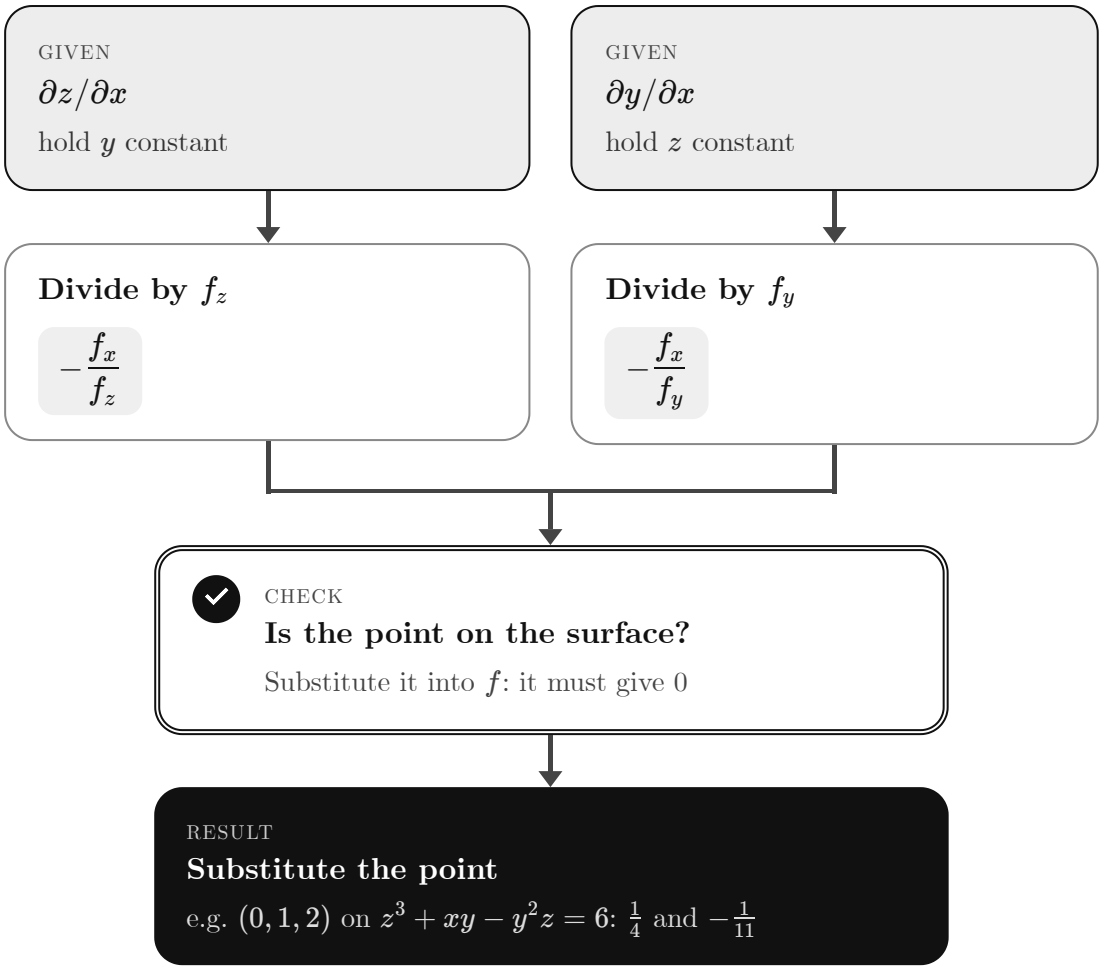
E7.1 · Implicit differentiation, three variables ESA ★★

$$f(x,y,z)=0: \quad \frac{\partial z}{\partial x}=-\frac{f_x}{f_z}, \quad \frac{\partial z}{\partial y}=-\frac{f_y}{f_z}\ (f_z\neq 0); \qquad \frac{\partial y}{\partial x}=-\frac{f_x}{f_y}\ (f_y\neq 0)$$

gives	Partial derivatives of the implicitly defined variable.
applies	Hold y constant for z_x; hold z constant for y_x.
note	Extends to any number of variables.

S004·6 · S003·26 · T004a 00:10:23 · T003b 00:09:29

Flowcharts



Exam trap. The lecture used $(1,0,2)$, which is not on the surface; always check the point first.

Figure F7.1. Partial derivatives from $f(x,y,z)=0$. ESA ★★

S004·6 · S004·9 · S003·26

Important points and why

1. For $f(x,y,z)=0$, x and y are independent, so dz alone has no meaning; find the partials z_x, z_y. ESA ★★ T004a 00:09:38
2. For $\partial z/\partial x$ treat y as constant; for $\partial y/\partial x$ treat z as constant. ESA ★★ S004·6 · S004·9 · T004a 00:19:00
3. The formula avoids separating z from a messy equation such as $x^3y^2+3xyz+\sin(e^z)=c$. ESA ★ T004a 00:10:55
4. Check that the given point satisfies $f=0$ before substituting. ESA ★★ ADDED FOR CLARITY S004·9 · T004a 00:18:39

Exam trap. The lecture works problem 3 at $(1,0,2)$ and gets 0 for both derivatives, but $(1,0,2)$ is not on $z^3+xy-y^2z=6$ (it gives 8). Use the slide's point $(0,1,2)$: $\partial y/\partial x = 1/4$, $\partial z/\partial x = -1/11$. ADDED FOR CLARITY S004·9 · T004a 00:20:47

Exam trap. Do not differentiate $-f_x/f_z$ as if z were a constant when you need second derivatives: z is still a function of x and y (W7.3). ADDED FOR CLARITY ESA Jan-24 s3 1d

Worked examples

W7.1 · $\partial y/\partial x$ and $\partial z/\partial x$ at $(0,1,2)$ — ESA ★★

GIVEN $z^3+xy-y^2z=6$ at $(0,1,2)$.

STEP 1 Check the point: $8+0-2=6$.

STEP 2 $f=z^3+xy-y^2z-6$: $f_x=y$, $f_y=x-2yz$, $f_z=3z^2-y^2$.

STEP 3 $\dfrac{\partial y}{\partial x}=-\dfrac{y}{x-2yz}=-\dfrac{1}{0-4}$; $\dfrac{\partial z}{\partial x}=-\dfrac{y}{3z^2-y^2}=-\dfrac{1}{12-1}$.

ANSWER $\boxed{\dfrac{\partial y}{\partial x}=\dfrac{1}{4},\quad \dfrac{\partial z}{\partial x}=-\dfrac{1}{11}}$

CHECK The lecture's point $(1,0,2)$ is not on the surface (see Exam trap).

S004·9 · T004a 00:18:39

W7.2 · $\partial z/\partial x$ and $\partial z/\partial y$ at the origin — ESA ★★

GIVEN $x^3+z^2+ye^{xz}+z\cos y=0$ at $(0,0,0)$.

STEP 1 $F_x=3x^2+zye^{xz}$, $F_y=e^{xz}-z\sin y$, $F_z=2z+xye^{xz}+\cos y$.

STEP 2 At the origin: $F_x=0$, $F_y=1$, $F_z=1$.

ANSWER $\boxed{\dfrac{\partial z}{\partial x}=0,\quad \dfrac{\partial z}{\partial y}=-1}$

CHECK The origin satisfies the equation.

S003·26 · T003b 00:09:15

W7.3 · Second derivative of an implicit function ESA ★★

GIVEN $x^x y^y z^z = c$. Show that at $x = y = z$, $\dfrac{\partial^2 z}{\partial x\,\partial y} = -[x\log(ex)]^{-1}$.

STEP 1 Take logs: $f = x\log x + y\log y + z\log z - \log c = 0$, so $f_x = 1 + \log x$, $f_y = 1 + \log y$, $f_z = 1 + \log z$.

STEP 2 $z_x = -\dfrac{1+\log x}{1+\log z}$, $z_y = -\dfrac{1+\log y}{1+\log z}$.

STEP 3 $z_{xy} = \dfrac{\partial}{\partial y} z_x = \dfrac{1+\log x}{(1+\log z)^2}\cdot\dfrac{1}{z}\,z_y$ (z depends on y).

STEP 4 At $x = y = z$: $z_y = -1$, so $z_{xy} = -\dfrac{1}{x(1+\log x)}$, and $1 + \log x = \log(ex)$.

ANSWER $\boxed{\dfrac{\partial^2 z}{\partial x\,\partial y} = -\dfrac{1}{x\log(ex)}}$

CHECK sympy residual 0.

ADDED FOR CLARITY ESA Jan-24 s3 1d · CW U1 P6

Past-paper questions

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q1.d · solved in W7.3

For the point on the surface $x^x y^y z^z = c$, show that at $x = y = z$, $\frac{\partial^2 z}{\partial x \partial y} = -[x\log(ex)]^{-1}$.

SHORT-NOTE RECAP

For $f(x,y,z) = 0$, z is a function of x and y: $\dfrac{\partial z}{\partial x} = -\dfrac{f_x}{f_z}$ (hold y), $\dfrac{\partial z}{\partial y} = -\dfrac{f_y}{f_z}$, and $\dfrac{\partial y}{\partial x} = -\dfrac{f_x}{f_y}$ (hold z), provided the denominators are non-zero. Check that the point lies on the surface. For second derivatives, remember that z still depends on x, y.

A Symbol glossary

Symbol	Name	Meaning	Appears in
$z = f(x, y)$	Function of two variables	The quantity being differentiated	E2.1, E3.1, E5.1
x, y	Intermediate variables	Depend on t (total derivative) or on u, v (chain rule)	E2.1, E5.1
t	Parameter / time	Single final variable in the total derivative; also a new variable in some chain-rule problems	E2.1, E2.2
u, v / s, t	New independent variables	Final variables of a composite function	E5.1
r, s, t	Named arguments	$r = 2x - 3y$, ... in slide 003·15 (not a radius there)	E5.2
r, θ	Polar coordinates	$x = r\cos\theta$, $y = r\sin\theta$	E5.4, E5.5
$\dfrac{dz}{dt}$	Total derivative	Ordinary derivative when z ends up depending on t alone	E2.1
$\dfrac{\partial z}{\partial u}$	Partial derivative of a composite	z depends on u and v	E5.1
$dx,\ dy,\ dz$	Differentials	Small (signed) changes; $dz = f_x\,dx + f_y\,dy$	E3.1, E3.2
f_x, f_y, f_z	Partial derivatives of f	Kept as symbols when f is unknown	E5.1, E6.1, E7.1
$f(x, y) = c$	Level curve / implicit relation	Defines y as a function of x	E6.1
$f(x, y, z) = 0$	Level surface	Defines z as a function of x, y	E7.1
$f'(r)$	Ordinary derivative of $f(r)$	u depends on x, y, z only through r	E5.3
m	Slope	dy/dx at the point; tangent $y - y_0 = m(x - x_0)$	E6.2
A, V, h, b, θ	Geometric quantities	Area, volume, height, base, included angle in rate problems	E4.1

B Sources, coverage and gaps

Source	Path	Use
Deck 003 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.*	28 slides; content on 4–27
Deck 004 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.*	18 slides; content on 3–17
Transcript 003 part I (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.*	26.5 min lecture
Transcript 003 part II (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.*	11.6 min problems
Transcript 004 part I (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.*	22.4 min lecture
Transcript 004 part II (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.*	11.7 min problems
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 9–18 (with answers)
ESA papers, UE22MA141A: Dec 2023, Jul 2023, May 2024	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE22MA141A/*.pdf	Q1 read; none in scope
ESA papers, UE23MA141A: Jan 2024 sets 1–4, June 2024	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE23MA141A/*.pdf	6 in-scope Q1 parts
Videos 003 and 004 (4 .mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/00[34]_*.mp4	Not opened: every gap was settled from the slides, transcripts and algebra

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. In-scope Q1 parts: five chain-rule questions (Jan 2024 set 1 Q1.d and set 2 Q1.d, 5 marks each; set 3 Q1.b, set 4 Q1.b and June 2024 Q1.a, 6 marks each) and one implicit-function question (Jan 2024 set 3 Q1.d, 6 marks). The total derivative and rates of change were not asked in these papers; they are computed in class-work

problems 9–13. No paper repeats another. Questions on Euler's theorem, Taylor series, maxima and Lagrange belong to later sessions and are left out.

Known gaps, conflicts and corrections

1. Slide 003·6 ($z = x^2y^3$, $x = t^2$, $y = 2t^3$) is wrong: it gives $\partial z/\partial x = 2t^{11}$, $\partial z/\partial y = 9t^{10}$ and $dz/dt = 60t^{12}$. The correct values are $16t^{11}$, $12t^{10}$ and $dz/dt = 104t^{12}$, confirmed by substituting first ($z = 8t^{13}$). The lecture (T003a 00:02:05) used a different example, $x = t$, $y = 2t^2$, giving $64t^7$.
2. Slide 003·10 prints $d\vartheta/dt = -25/(300\sqrt{3})$ "$= 0.04831$", and the lecture repeats 0.04831. The correct value is $-\sqrt{3}/36 \approx -0.0481$ rad/s, which the slide's own last line states.
3. Slide 003·8 writes "dy = 0.03", but the calculation correctly uses $dy = -0.03$ (from -3 to -3.03).
4. Slide 003·16 shows $\partial v/\partial y = -3\,\partial v/\partial r + \partial v/\partial s$; the coefficient of $\partial v/\partial s$ should be 3 (since $s = 3y - 4z$). The final result $6v_x + 4v_y + 3v_z = 0$ is right.
5. Slide 003·7 calls $x^2/16 - y^2/9 = z$ a paraboloid and the lecture says "like a cup". With the minus sign it is a hyperbolic paraboloid (a saddle). The numbers are unaffected.
6. Slide 004·4 prints $dy/dx = -f_y/f_x$ for a level curve. It should be $-f_x/f_y$, as slide 004·5's branch diagram and slide 004·6 show.
7. Slide 004·3's numerator is hard to read in the rendering. The correct dy/dx for $x^3y + 3xy^2 = 3$ is $-3y(x^2 + y)/(x(x^2 + 6y))$.
8. In the lecture (T004a 00:18:39) problem 3 of deck 004 is worked at the point (1, 0, 2), giving 0 for both derivatives. That point is not on $z^3 + xy - y^2z = 6$ (it gives 8). The slide's point (0, 1, 2) is on the surface and gives $\partial y/\partial x = 1/4$, $\partial z/\partial x = -1/11$.
9. The lecture misspeaks some intermediate values (e.g. "S X is just minus 4", T003a 00:23:23); the slides' values $r_x = 2$, $s_x = 0$, $t_x = -2$, ... are correct and are used here.
10. Transcript recognition errors were corrected silently: "x pi" → xy, "cos by" → cos y, "second square" → $\sec^2$, "phi dy by dx" → find dy/dx, "Gosteep" → cos ϑ, "x incre" → x increases.

Coverage

Source	Citations	Slides cited (slide × citations)
S003	81	4×4, 5×10, 6×2, 7×6, 8×7, 9×6, 10×4, 11×3, 12×2, 13×6, 14×3, 15×4, 16×3, 17×3, 18×2, 19×2, 20×1, 21×2, 22×1, 23×1, 24×1, 25×2, 26×3, 27×3
S004	60	3×6, 4×11, 5×7, 6×8, 7×4, 8×3, 9×5, 10×2, 11×1, 12×2, 13×3, 14×2, 15×3, 16×1, 17×2
CW	15	
ESA	9	
T003a	48	
T003b	16	
T004a	35	
T004b	14	

Slides 003·1–3, 003·28, 004·1–2 and 004·18 are title, topic-list and closing slides with no content.